International Baccalaureate®
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Computer Science

Higher and Standard level

Specimen papers 1 and 2

For first examinations in 2027

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Computer Science higher level and standard level: Case study

Diploma Programme
Programme du diplôme
Programa del Diploma

Computer science
Higher level
Paper 1

Specimen paper

Candidate session number

2 hours

Instructions for candidates

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- A clean copy of the **computer science case study** is required for this examination paper.
- Read the case study carefully.
- Section A: answer all the questions.
- Section B: answer the question.
- Answers must be written within the answer boxes provided.
- The maximum mark for this examination paper is **[80 marks]**.

16 pages

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Section A

Answer **all** questions from this section. Answers must be written within the answer boxes provided.

1. A team of programmers use platform as a service (PaaS) via their laptops to develop a software application.

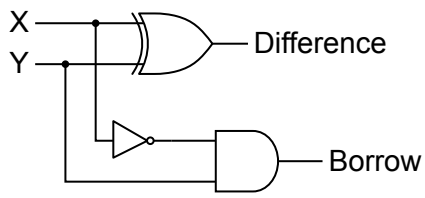
(a) Outline the function of an accumulator. [2]

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(b) State the binary equivalent of the hexadecimal number AC. [1]

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(c) Construct a truth table for the following circuit: [2]



X	Y	Difference	Borrow

(This question continues on the following page)

(Question 1 continued)

(d) Outline the role of accounting in an operating system. [2]

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(e) Outline why platform as a service (PaaS) is a suitable environment in which to build applications. [2]

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(f) Explain why the execution speed of compiled code is faster than that of interpreted code. [4]

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2. A hospital has a local area network (LAN) connected to the internet to provide access to cloud services.

(a) State **one** network device used to connect the local area network (LAN) to the internet. [1]

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The TCP/IP model was developed to facilitate reliable and robust communication over a network.

(b) Describe the function of the network interface layer of the TCP/IP model. [2]

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A router uses a fiber-optic cable to connect directly to an internet service provider's equipment.

(c) List **three** advantages of using a fiber-optic cable instead of a twisted-pair cable. [3]

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(This question continues on the following page)

(Question 2 continued)

(d) Explain how a switch routes data packets across a network. [4]

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The hospital deploys a hardware firewall incorporating network address translation (NAT).

(e) Explain **one** function of network address translation (NAT) in a firewall. [4]

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3. A hospital stores patients' data in a relational database.

The database view, VISITS, is an extract showing the dates patients visited the hospital for an appointment.

The VisitDate field represents the date of each patient's appointment.

PatID	PatientName	PatientDOB	DocID	Doctor	VisitDate	Treatment
1	Samuel Moody	2007/06/11	1	Jill Smith	2025/01/11	Bone scan
1	Samuel Moody	2007/06/11	2	Norayr Lomidze	2025/01/12	Physical therapy
2	Lin Kobayashi	1984/02/24	1	Jill Smith	2025/01/12	Physical therapy
3	Yeojoo Kim	1999/08/18	3	Phoebe Drakos	2025/01/13	Radiation therapy
3	Yeojoo Kim	1999/08/18	3	Phoebe Drakos	2025/01/14	Therapy part 1

(a) Construct an SQL SELECT query to list the names of patients whose treatment includes the word "therapy".

A patient's name should only appear once. [3]

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(Question 3 continued)

The database view, VISITS, can be represented by the following notation:

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VISITS (PatID, PatientName, PatientDOB, DocID, Doctor,
VisitDate, Treatment)
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(b) Construct the database in third normal form (3NF) for the three entities in the VISITS view extract. [5]

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The hospital has transitioned its database system to a distributed model. Atomicity can be applied across multiple nodes when transactions are executed.

(c) Describe how atomicity manages transactions across multiple nodes. [3]

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(This question continues on page 9)

Please **do not** write on this page.

Answers written on this page
will not be marked.

(Question 3 continued)

The hospital's research laboratory tracks virus outbreaks. A spatial database stores the addresses of patients who have contracted a virus.

(d) Outline why a spatial database is used to store the addresses of patients who have contracted a virus. [3]

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4. A hospital uses a deep learning convolutional neural network (CNN) to analyse chest X-rays and predict the likelihood of a patient developing pneumonia.

(a) State **one** of the layers in a convolutional neural network (CNN). [1]

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(b) Identify **one** reason data cleaning is carried out before training a convolutional neural network (CNN). [1]

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(c) Outline **one** reason a graphics processing unit (GPU) is recommended when training a deep learning convolutional neural network (CNN) model using a large dataset. [2]

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(d) Describe how the deep learning convolutional neural network (CNN) model could be evaluated using an F1 score. [3]

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(Question 4 continued)

The hospital purchased a self-learning robotic cleaner that uses reinforcement learning (RL).

(e) Justify why a robotic cleaner would use reinforcement learning (RL) instead of supervised learning (SL). [3]

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A deep learning algorithm was implemented to automatically monitor the hospital's social media sites. The model uses natural language processing to classify comments into categories like "compliments" and "complaints".

Members of the public are asked to remove any comments that are not supported by evidence. However, the hospital's CEO is concerned about the ethical issues of doing this, as it may be considered online harassment.

(f) Discuss whether the hospital should ask members of the public to retract their comments. [5]

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Section B

Answer the following question, which refers to the case study. Answers must be written within the answer boxes provided.

5. Ransomware and rootkit are types of malware.

(a) (i) Identify **two other** types of malware. [2]

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A phishing attack caused ransomware to be installed on TaylorMade Healthcare's network, resulting in financial loss.

(ii) Outline **one other** potential consequence of a phishing attack on a hospital. [2]

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(Question 5 continued)

A bring your own device (BYOD) policy allows employees to use their own devices to access data on TaylorMade Healthcare's local area network (LAN) via the internet.

(b) Outline **two** strategies that TaylorMade Healthcare could use to maintain security when employees use their own devices. [4]

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(c) Outline **two** advantages of a bring your own device (BYOD) approach compared with the hospital providing devices to employees. [4]

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(This question continues on the following page)

(Question 5 continued)

TaylorMade Healthcare identified several challenges to build cyber resilience.

(d) To what extent can institutions like TaylorMade Healthcare prevent and respond to cybersecurity attacks? [12]

Diploma Programme
Programme du diplôme
Programa del Diploma

Markscheme

Specimen paper

Computer science

Higher level

Paper 1

17 pages

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General marking instructions

1. Follow the markscheme provided, award only whole marks and mark only in **RED**.

2. Make sure that the question you are about to mark is highlighted in the mark panel on the right-hand side of the screen.

3. Where a mark is awarded, a tick/check (✓) **must** be placed in the text at the **precise point** where it becomes clear that the candidate deserves the mark. **One tick to be shown for each mark awarded**.

4. Sometimes, careful consideration is required to decide whether or not to award a mark. In these cases use RM™ Assessor annotations to support your decision. You are encouraged to write comments where it helps clarity, especially for re-marking purposes. Use a text box for these additional comments. It should be remembered that the script may be returned to the candidate.

5. Personal codes/notations are unacceptable.

6. Where an answer to a part question is worth no marks but the candidate has attempted the part question, enter a zero in the mark panel on the right-hand side of the screen. Where an answer to a part question is worth no marks because the candidate has not attempted the part question, enter an "NR" in the mark panel on the right-hand side of the screen.

7. Please ensure you check all scanned pages. The candidate may have answered more than one option.

8. **If a candidate has attempted more than one Option** within a paper, mark all the candidate's work. RM™ Assessor will only award the marks for the higher scoring Option. Once all the work the candidate has attempted has been marked, please click "COMPLETE"; all the other questions from the other Options will auto complete to "NR" for "no response".

9. Ensure that you have viewed **every** page including any additional sheets. Please ensure that you stamp "SEEN" on any page that contains no other annotation.

10. A mark should not be awarded where there is contradiction within an answer. Make a comment to this effect using a text box or the "CON" stamp.

Subject details: Computer Science HL Paper 1 markscheme

Mark allocation

Section A: Candidates are required to answer **all** questions. Total 56 marks.
Section B: Candidates are required to answer **all** questions. Total 24 marks.
Maximum total = 80 marks.

General

A markscheme often has more specific points worthy of a mark than the total allows. This is intentional. Do not award more than the maximum marks allowed for that part of a question.

When deciding upon alternative answers by candidates to those given in the markscheme, consider the following points:

Each statement worth one point has a separate line and the end is signified by means of a semi-colon (;).

An alternative answer or wording is indicated in the markscheme by a "/"; either wording can be accepted.

Words in (…) in the markscheme are not necessary to gain the mark.

If the candidate's answer has the same meaning or can be clearly interpreted as being the same as that in the markscheme then award the mark.

Mark positively. Give candidates credit for what they have achieved and for what they have got correct, rather than penalizing them for what they have not achieved or what they have got wrong.

Remember that many candidates are writing in a second language; be forgiving of minor linguistic slips. In this subject effective communication is more important than grammatical accuracy.

Occasionally, a part of a question may require a calculation whose answer is required for subsequent parts. If an error is made in the first part then it should be penalized. However, if the incorrect answer is used correctly in subsequent parts then **follow through** marks should be awarded. Indicate this with "**FT**".

General guidance

Issue	Guidance
Answering more than the quantity of responses prescribed in the questions	In the case of an "identify" question, read all answers and mark positively up to the maximum marks. Disregard incorrect answers. In the case of a "describe" question, which asks for a certain number of facts *eg* "describe two kinds", mark the first two correct answers. This could include two descriptions, one description and one identification, or two identifications. In the case of an "explain" question, which asks for a specified number of explanations *eg* "explain two reasons …", mark the first two correct answers. This could include two full explanations, one explanation, one partial explanation *etc*.

Section A

1. (a) *Award **[2]** max*

To store intermediate arithmetic and logic results/hold operand data;
…to reduce the number of memory accesses (by holding intermediate results);

Acts as a temporary storage location;
…which stores the values and results from arithmetic and logical operations;

Holds data or the results of an operation;
…during the processing cycles;

(b) *Award **[1]** max*

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(c) *Award **[2]** max*

*Award **[1]** for each correct column (Difference, Borrower)*

X	Y	Difference	Borrower
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

(d) *Award **[2]** max*

Tracks resource usage;
Monitors and logs the consumption of resources like CPU, memory, and disk by various processes and users;

Assesses system performance;
Collects data to identify system bottlenecks, under-utilized resources, or areas for potential optimization;

Assigns costs based on usage;
In shared computing environments, users may be billed based on the resources they consume;

Facilitates auditing;
Helps in detecting abnormal behaviors or potential security breaches;

(e) *Award **[2]** max*

*Award **[1]** per point up to **[2]** max*

Offers a range of pre-built components and libraries to accelerate development;
Developers can concentrate on coding rather than managing infrastructure;
Cloud hosted to allow multiple team members to collaborate on application development;
PaaS often comes with built-in tools for development, testing, integration, and deployment;
PaaS providers often have strong security protocols to protect the application and data;
PaaS solutions are designed to be scalable, allowing for easy expansion;

(f) *Award **[4]** max*

*Award **[1]** per point up to **[4]** max*

Pre-translation: Compiled code runs faster because it has been pre-translated into machine code, while interpreted code is slower due to the on-the-fly interpretation;

Runtime overhead: Compiled code is pre-compiled, so there's no runtime overhead associated with translating source code to machine code, whereas an interpreter must parse, interpret, and translate each line or block of code in real time;

Code optimization: Compilers can take advantage of the architecture's specific capabilities (e.g., optimizing for loops, tasks), whereas interpreter does not have the full view of the program;

Resource utilization: Compilers are more efficient with CPU and memory resources during execution, whereas interpreters consume more CPU cycles and memory during execution (due to the overhead of interpreting code line by line);

Cache utilization: Compiled code has better cache utilization due to the compilation optimizations, whereas the interpreter itself adds an extra layer of instructions that need to be processed, leading to cache misses and slower execution.

2. (a) *Award **[1]** max*

Router/Modem/Gateway

(b) *Award **[2]** max*

*Award **[1]** for point, **[1]** for expansion*

The network interface layer is responsible for the physical transmission of data;
...including the layout of bits into packets, frames, or other formats;

The network interface layer manages interactions between devices/hardware;
...and the network medium itself, such as Ethernet cable or Wi-Fi;

The network interface layer handles error notification;
...Identifying and reporting any issues or faults in the data transmission process (e.g. lost packets or a disconnected network cable);

The network interface layer determines the network topology (e.g., star, bus, ring, hybrid);
...which impacts network operations (e.g., scalability, fault-tolerance, and data transmission efficiency).

The network interface layer manages physical addressing;
...including assignment and identification of devices (e.g., MAC addresses).

(c) *Award **[3]** max*

*Award **[1]** per point*

Transmission speed/higher bandwidth;
Lower latency compared to twisted-pair cables;
Transmits signals over greater distances than twisted-pair cables;
Immune to electromagnetic interference (EMI) and radio-frequency interference (RFI);
Better security against unauthorized access or data tapping;
Better scalability as data transmission needs grow;
Generally more durable than twisted-pair cables;
Longer lifespan and lower maintenance requirements;

(d) *Award **[4]** max*

*Award **[1]** per point*

***MAC address learning*:** A switch learns the device's MAC address (from the incoming packet's source address)/a switch maintains a MAC address table;
…and associates the MAC address with the physical port through which the packet entered;

***Packet examination*:** destination MAC address is shown in the frame/packet;
…and compared with the switch's MAC address table to find the port;

Selective forwarding: The switch determines the appropriate port to forward the packet/frame based on its destination MAC address;
If the destination MAC address is unknown the switch broadcasts the packet (frame) to all ports;

Possible candidate response worth full marks
The switch saves the arrival port/sender mac address extracted from packet;
The destination MAC address is extracted from the packet;
And the port is looked up in a table based on the MAC address;
The switch then plots the path to the destination port.

Note: either frame or packet is acceptable for candidates' answers.

(e) *Award **[4]** max*

*Award **[1]** for each point from the same section*

NAT translates private IP addresses of devices within a local network to a public IP address;
External entities see only the public IP address of the network;
NAT makes it more difficult for attackers to directly reach internal systems;
The extra layer of obscurity enhances the security of the network;

NAT and firewalls often work together;
Firewall inspects and filters incoming and outgoing traffic based on security rules;
NAT allows network administrators to control how traffic is routed;
NAT can be configured to forward specific types of traffic to specific internal devices;

Not only does NAT add an extra layer of security;
NAT enables multiple devices on a local network to share a single public IP address.
Useful given the limited availability of IPv4 addresses;
Enables the network to have more devices than they have IPv4 IP addresses;

3. (a) *Award **[3]** max*

*Award **[1]** for SELECT DISTINCT or GROUP BY*

*Award **[1]** for using LIKE*

*Award **[1]** for using the wildcard (any wildcard accepted) and therapy*

```
SELECT DISTINCT PatientName FROM VISITS
WHERE Treatment LIKE '%therapy%';
```

Alternative solution using group by:

```
SELECT PatientName FROM VISITS
WHERE Treatment LIKE '%therapy%'
GROUP BY PatientName
```

Alternative solution using a subquery and group by:

```
SELECT PatientName
FROM (
    SELECT PatientName
    FROM VISITS
    WHERE Treatment LIKE '%therapy%'
) AS subquery
GROUP BY PatientName;
```

(b) *Award **[5]** max*

*Award **[1]** for the PATIENT table with correct Primary Key*

*Award **[1]** for the DOCTOR table with correct Primary Key*

*Award **[1]** for PatID foreign key in the VISIT table*

*Award **[1]** for DocID foreign key in the VISIT table*

*Award **[1]** for VisitID or PatID/DocID/VisitDate composite key in the VISIT table*

```
PATIENT(PatID, PatientName, PatientDOB)
DOCTOR(DocID, Doctor)
VISIT(VisitID, PatID, DocID, VistDate, Treatment)
```

(c) *Award **[3]** max*

*Award **[1]** per point*

Atomicity assures that a transaction is treated as a single, indivisible unit;
A transaction must either be fully completed on all nodes or not at all;
Transaction failure on one node triggers a rollback of the entire transaction across all nodes;
A commit will make sure that the updates which happened on one node because of the transaction are persisted in all nodes.

(d) *Award **[3]** max*

*Award **[1]** per point*

Spatial DBs are designed to store geographic data such as addresses;
Spatial DBs can be queried to identify clusters of virus patients;
Spatial DBs can create visual representations (maps) to see patterns;
Spatial DBs can support advanced spatial analysis, calculating distances between patient addresses and hospitals or identifying the cause e.g. nearby research lab;
Enables resource allocation (medical teams can be dispatched to the region);
Stored address data enables location-based contact tracing/highlights the area to quarantine/notify neighbours of the outbreak;

4. (a) *Award **[1]** max*

Input layer;
Output layer;
Convolution layer/conv layer;
Pooling layer/subsampling layer/down-sampling layer;
Fully-connected layer/FC layer;
Normalization layer;
Activation layer.

(b) *Award **[1]** max*

To improve model accuracy/ensure the model is not trained on poor quality data;
To ensure Inaccuracies/errors/inconsistencies in the data doesn't lead to false assumptions;
To removing duplicate data/irrelevant data points;
Reduce the training time/computational resources required;
To handle missing values by using imputation (e.g. inserting mean) or removing;
Ensures the CNN doesn't make predictions on incomplete data/have gaps in its knowledge;
To provide uniformity of data/ensure all images are the same size and format;
To ensure the model isn't giving undue importance to some features over others;
To balance the dataset (undersampling or oversampling);
To ensure the dataset doesn't have an uneven distribution of classes/become biased towards the majority class;
To avoid overfitting;
To remove outliers or extreme values in the training data that can cause overfitting.

(c) *Award **[2]** max*

*Award **[1]** per point from one section*

Deep learning involves many matrix operations (multiplications and convolutions);
GPUs have many smaller cores designed explicitly for parallel operations;

Large datasets increase the computational demand;
Training on a CPU only would be too time-consuming/GPUs speed up the training time;

Converges by minimizing the cost function (e.g., batch gradient descent) lending itself to simultaneous operations;
GPUs provide parallel processing, leading to faster convergence;

Accept any reasonable machine learning process that requires parallelization

(d) *Award **[3]** max*

*Award **[1]** per point*

Calculate the F1 score as the harmonic mean of precision and recall;
A low F1 score implies inadequate performance of the CNN model;
This may require training adjustments or data rebalancing;
It is an iterative process of adjusting and testing to improve the F1 score.

Alternative answer
Produce a confusion matrix (TP, TN, FP, FN);
Calculate precision using TP / (TP + FP);
Calculate recall/sensitivity using TP / (TP + FN);
Calculate F1 score = 2 * (Precision * Recall) / (Precision + Recall).

Note to examiners:
TP is true positive (correctly identified cases of pneumonia);
TN is true negative (correctly identified cases without pneumonia);
FP is false positive (wrongly identified cases of having pneumonia);
FN is false negative (cases of pneumonia that were not identified).

(e) *Award **[3]** max*

*Award **[1]** for identifying an **issue***

*Award **[1]** for expanding on the issue/stating the **context***

*Award **[1]** for a further **evidence** highlighting the choice of RL over SL*

RL works well in **dynamic environments** where layouts change/hospitals are dynamic environments (beds move, machinery is moved);
RL involves the cleaner (agent) interacting with the environment (hospital), receiving continuous feedback based on its actions (movement decisions);
RL allows the robot cleaner to update its learned policy through rewards and punishments;
SL is trained in advance on a static dataset so would struggle to adapt to untrained layouts/changing environments;
SL requires accurate labelled datasets, which is impractical for every possible situation or condition in a hospital;

Alternative answer:
RL allows for **real-time autonomous learning** (without interventions);
The robot (agent) uses RL to interact with the environment making decisions;
The feedback loop in RL (perform action, get reward, update policy) ensures the robot continuously improves its strategy based on real-time outcomes;
SL is trained before use and does not learn in real-time;
Changes in the environment would require external intervention (retraining);
A dataset with all possible layouts and actions is implausible;

Alternative answer:
RL is goal driven/has an **ultimate goal** of cleanliness;
RL is designed to allow the cleaner to discover an optimal sequence of actions using trial-and-error;
RL, unlike SL, can handle the situation where there are delayed rewards;
SL is not designed to find the best sequence of actions;
SL is designed to predict outcomes based on provided input;
SL is suited to classification and regression tasks, not optimization tasks.

(f) *Award **[5]** max*

The following points might be considered:

- Definition of online harassment - any form of bullying or intimidation that takes place through digital mediums
- Consideration of the context because much depends on the comments
- The sensitivity shown by hospital staff – only online harassment when they harm, threaten, or are offensive. They may only be asking for retractions in extreme circumstances
- The method of contacting public members (online, email, phone)
- How the hospital obtained members contact details (right to privacy)
- Freedom of speech and protection from unsubstantiated comments
- Potential impact on the public's trust in the hospital
- Alternative suggestions to the current practice to alleviate ethical concerns
- Legal implications such as GDPR
- Impacts of online harassment can be severe, leading to emotional distress, anxiety, depression, and even self-harm or suicide in extreme cases
- Is it ethical to use machine learning in this situation
- Consideration of the training data used for sentiment analysis
- The accuracy and potential bias (e.g. ethnicity, socio-economic status, gender) of the model

Marks	Level descriptor
No marks	No knowledge or understanding of the relevant issues and concepts.
1–2 marks	Limited knowledge and understanding of harassment. The answer may be little more than a list of points. (General)
3–4 marks	Detailed knowledge that **links** harassment to the scenario; Fairly balanced analysis. (In context)
5 marks	An appropriate conclusion, linked to the analysis.

Section B

5. (a) (i) *Award **[1]** per line up to **[2]** max*

Adware
Backdoor
Bots/botnet
Crypto-jacking
Drive-by download
Exploit kit
Fileless malware
Keylogger
Polymorphic malware
Spyware
Trojan horse
Virus
Worm
Zero-day exploit

Accept other examples of legitimate malware

(ii) *Award **[2] max***

*Award **[1]** for identifying issue and **[1]** for expansion*

Reputational damage;
Clients may lose confidence in a company that has a breach of security;
For example, parents may withdraw their students from a school;

Financial fraud for patients;
Patients' credit card numbers/banking data/passwords may result in funds being removed;

Theft of data;
Patients' data is used unethically (e.g., insurance company refusing to insure patients with potential pre-existing medical conditions);

Identity theft;
Having your identity stolen can cause psychological damage; e.g., a person may steal a person's online profile and insult others, causing emotional distress;

Data loss causing disruption to hospital services;
Phishing attacks are often designed to steal sensitive data; e.g., doctors lose access to patients' records and can't treat them;

Legal consequences;
Company may be sued for leaking sensitive information by patients/companies/government/ombudsman and lose their license to practice medicine;

Additional malware installed;
Creates vulnerability for the additional malware to cause further disruption.

(b) *Award **[4]** max*

*Award **[1]** for identifying strategy and **[1]** for expansion*

Implement **device authentication** protocols;
Registering devices, require complex passwords, two-factor authentication, certificate authentication, or biometric data to access the network;

Implement **access control** protocols;
Restricting the user access based on their role;

Use **device management software**/mobile device management (MDM);
Allows IT to manage, monitor, and secure employees' mobile devices;

Require **security updates**/patches be installed;
Devices without the updates/patches will trigger an alert to IT admin;

Install **anti-malware software** on all devices;
Ensure it is kept up to date/push updates to the users' devices;

Remote wipe software;
Install software to delete company data on lost devices remotely;
End-user technical agreement agreeing to device use;
Employee signs as part of their contract;

User training on cybersecurity;
To reduce the risk of cybersecurity breaches;

*Mark as **[2]** + **[2]***

(c) *Award **[4]** max*

*Award **[1]** for advantage, and **[1]** for expansion*

Cost efficiency: Reduces initial costs on purchasing hardware;
Maintenance, repair, and replacement costs shifts to employ;

Familiarity: Users are more familiar with their own devices;
Leading to increased productivity/satisfaction and reduced training needs;

Personal choice: Employees are empowered to choose their own devices;
Gives them autonomy/sense of ownership/enables remote working;

*Mark as **[2] + [2]***

(d) *Award **[12]** max*

Candidates should discuss prevention and response, expressing their opinions on the effectiveness of the strategies. The best responses will demonstrate research and cite real-world related events and/or experts in the field.

Prevention
Implementing robust security protocols: TaylorMade Healthcare can establish stringent security protocols, including firewalls, encryption, email filtering, and regular software updates. These measures can prevent many common attacks, but candidates should be aware that security measures quickly become outdated.

Access control and authentication: Implement strict access control policies, ensuring employees have access only to the data necessary for their role. Use strong authentication methods like two-factor authentication.

User awareness training: Educating staff about potential threats, such as phishing scams and ransomware attacks, can prevent many attacks. Candidates should be aware that human error remains a significant challenge. TaylorMade Healthcare should have clearly defined procedures and protocols for new staff and departing staff. No matter how much training is provided, it is virtually impossible to eliminate all human errors that might lead to a security breach.

Network segmentation and access control: This strategy can limit the damage if a breach does occur by preventing hackers from accessing the entire network. Candidates should be aware that this strategy may not prevent the initial breach and can be complex to implement and manage.

Update and patch management: Ensure all software, including operating systems and applications, are kept up to date with the latest security patches.

Regular security audits and assessments: Conduct regular security audits and risk assessments to identify and address vulnerabilities.

Monitoring and logging: Continuously monitor network activity and maintain logs to detect and investigate suspicious activities.

Physical security: Enhance physical security measures to prevent unauthorized access to servers and other critical IT infrastructure.

Vendor risk management: Assess and manage the cybersecurity risks associated with third-party vendors and service providers.

Cybersecurity insurance: Consider purchasing cybersecurity insurance to mitigate financial risks associated with data breaches.

Collaboration with law enforcement and cybersecurity agencies: Work with law enforcement and cybersecurity agencies for threat intelligence sharing and guidance on emerging threats and best practices.

Response
Data backup and disaster recovery plans to limit the damage caused by an attack and ensure that services can quickly resume after a breach. Candidates should be aware that implementing these measures requires significant resources and that recovering from a breach can still be a lengthy and expensive process.

Third-party services: Cybersecurity consultants and managed security service providers can help respond to breaches quickly and effectively. Candidates may consider whether TaylorMade Healthcare can afford this approach because these services can be expensive.

Failover systems: These systems ensure that critical services remain available even if the primary system fails. However, failover systems require significant investment and ongoing maintenance.

Conclusion
A reasoned conclusion that expresses an opinion supported by evidence.

Marks	Level descriptor
No marks	• No knowledge or understanding of the relevant issues and concepts. • No use of appropriate terminology.
Basic 1–3 marks	• Minimal knowledge and understanding of the relevant issues or concepts. • Minimal use of appropriate terminology. • The answer may be little more than a list. • No reference is made to the information in the case study or independent research.
Adequate 4–6 marks	• A descriptive response with limited knowledge and/or understanding of the relevant issues or concepts. • A limited use of appropriate terminology. • There is limited evidence of analysis. • There is evidence that limited research has been undertaken.
Competent 7–9 marks	• A response with knowledge and understanding of the related issues and/or concepts. • A response that uses terminology appropriately in places. • There is some evidence of analysis. • There is evidence that research has been undertaken.
Proficient 10–12 marks	• A response with a detailed knowledge and clear understanding of computer science. • A response that uses terminology appropriately throughout. • There is competent and balanced analysis. • Conclusions are drawn that are linked to the analysis. • There is clear evidence that extensive research has been undertaken.

Diploma Programme
Programme du diplôme
Programa del Diploma

Computer science
Higher level
Paper 2

Specimen paper

2 hours

Instructions to candidates

- Do not open this examination paper until instructed to do so.
- Answer all of the questions from **one** of the options.
- The maximum mark for this examination paper is **[80 marks]**.

Option	Questions
Option A — Java	1 – 6
Option B — Python	7 – 12

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Option A — Java

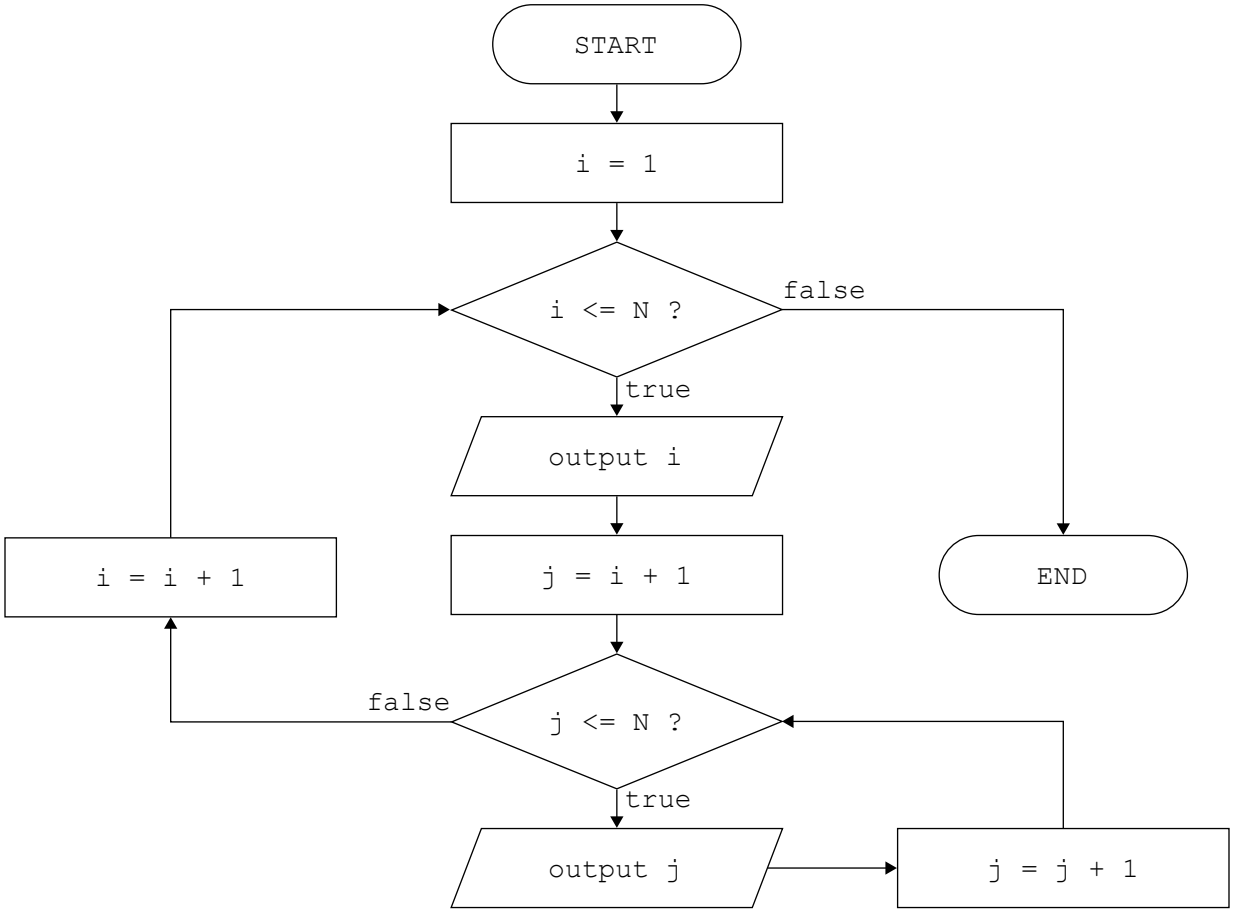
Answer **all** questions. All code that forms part of your responses must be written in Java.

1. Algorithms can be represented by flowcharts.

(a) State **one** reason why a flowchart could be used to represent an algorithm. [1]

Consider the flowchart shown in **Figure 1**.

Figure 1: Flowchart for an algorithm



(b) Copy and complete the trace table formed by the processing of this flowchart, where `N = 3`. [3]

i	i <= N?	j	j <= N ?	output
1				

(Option A continues on the following page)

(Option A, question 1 continued)

Big O notation is used to indicate the time complexity of an algorithm.

(c) State the Big O notation for the algorithm shown in **Figure 1**. [1]

The text file, employees, contains the following information about each employee in a company:

- Surname
- First name
- Role

Each item of data is stored in a new line. Details of the first three employees are shown:

Roberts
John
Sales
Saied
Abdullah
Manager
Sanchez
Pedro
IT manager
…

An algorithm reads the data from the file, employees, line by line. The algorithm writes only the surname of each employee into a new text file, surnames. Both files are currently closed.

(d) Describe the steps that would be followed by this algorithm. [5]

(Option A continues on page 5)

Blank page

(Option A, question 1 continued)

Stacks are a data structure frequently used in computer science to represent real-world scenarios.

Table 1 shows the fundamental operations that operate on a stack.

Table 1: Fundamental operations that operate on a stack

`push()`	Inserts an element onto the top of the stack.
`pop()`	Removes an element from the top of the stack.
`peek()`	Returns the top element of the stack without removing it from the stack.
`isEmpty()`	Returns true if the stack is empty, else it returns false.

(e) Sketch the stack resulting from the following operations.

```
push(1)
push(2)
push(9)
peek()
pop()
push(8)
push(10)
pop()
push(6)
```

You should clearly indicate the top of the stack. [2]

A new stack has been created and filled with elements.

(f) Using only the stack operations shown in **Table 1**, describe the steps that will locate **and** print only the **first** element that was pushed onto the new stack. [3]

(Option A continues on the following page)

(Option A continued)

2. A theatre stores the names and email addresses of its clients in two separate one-dimensional (1D) arrays, where each client's name and email address have corresponding indices. The two arrays are shown in **Table 2**.

Table 2: Name and email arrays

Name array

Index	Name
0	Maria Gonzalez
1	Olga Sarkozky
2	Julia Brown
3	…

Email array

Index	Email address
0	magonzalez24@example.net
1	osarkozky12@one2seven.org
2	jbrown23@example.com
3	…

(a) Explain why arrays have been chosen to hold this data instead of separate variables. [3]

A method has been constructed that will change each name so that the surname appears before the first name. For example, "Maria Gonzalez" will become "Gonzalez Maria".

(b) Describe the steps involved in this method.

Assume that there is only one space between each first name and surname. [4]

Consider the following two-dimensional (2D) array, `MAT`.

```
int[][] MAT = {{2, 7, 1}, {8, 9 ,0}}
```

(c) State the code that will identify the data item `0` in `MAT`. [1]

(Option A continues on the following page)

(Option A, question 2 continued)

The following code operates on `MAT`:

```
int[][] MAT = {{2, 7, 1}, {8, 9 ,0}};
int a = 0;
int b = 0;
for (int row = 0; row < 2; row++)
{
    for (int col = 0; col < 3; col++)
    {
        a = a + 1;
        int item = MAT[row][col];
        if (item%2 != 0)
            b = b + item;
    }
}
System.out.println ("The value of a is : " + a);
System.out.println ("The value of b is : " + b);
```

(d) State the output from this code. [2]

The theatre employs a 2D array to indicate the status of each seat for a performance. `true` indicates that the seat has been sold for that performance. `false` indicates that it has not been sold.

The 2D array `seats` is used to represent the seating plan in the theatre. The array contains 40 rows and 30 columns.

For example, `seats[2][3] = false` shows that the seat in row 2, column 3 of the 2D array, `seats`, has not been sold.

After each performance, the sales data is analysed with a program that uses the method `least()`. This accepts the 2D array seats and outputs the row for which the lowest number of seats were sold.

(e) Construct the method `least()`.

Assume only one row matches this condition. [5]

(Option A continues on the following page)

(Option A continued)

3. An airline uses object-oriented programming (OOP) to manage its booking system. This includes the class `Passenger`, which is partially shown.

```
public class Passenger{

    private String passengerName;
    private String email;
    private double ticketPrice;  // the price is determined when each
                                 // ticket is purchased

    public Passenger(String name, String email, double price)
    {
        this.passengerName = name;
        this.email = email;
        this.ticketPrice = price;
    }

    public String getPassengerName()
    {
        return passengerName;
    }
}
```

(a) (i) List the operations carried out when the following code is run:

```
Passenger p1 = new Passenger("Conejo Jorge",
"st34@one2seven.org", 240);
```

[3]

(ii) Outline **two** consequences of the use of the modifier `private` in this class. [2]

(Option A continues on the following page)

(Option A, question 3 continued)

The data of each flight are stored in a `Flight` object. This includes details of the passengers stored in an array of `Passenger` objects.

The `Flight` class is partially shown:

```
public class Flight{
    public int seatsBooked;   // incremented whenever a ticket is bought
    public int maxPassengers; // maximum number of passengers for this flight
    public double initialPrice; // initial price of a ticket on this flight
    public Passenger[] passengers; // array of Passenger objects

    public Flight(int mp, double ip)
    {
        this.seatsBooked = 0;
        this.maxPassengers = mp;
        this.initialPrice = ip;
        this.passengers = new Passenger[maxPassengers];
    }

    updatedTicketPrice() // code missing

    addPassenger()       // code missing

    selectSort()         // code missing
}
```

Multiple classes have a variety of relationships, including aggregation and composition.

(b) Outline the relationship between the `Passenger` and `Flight` classes. [2]

(Option A continues on the following page)

(Option A, question 3 continued)

The initial price of a ticket for each flight (`initialPrice`) is the value set by the airline before any tickets have been sold for that flight. However, the actual ticket price that the passenger pays (`ticketPrice`) will change depending on the number of tickets already sold.

The airline determines the actual price of a ticket as follows:
- It is equal to the initial price if less than 10 % of tickets have been sold.
- It is 1.5 times the initial price if 10 %–50 % of tickets have been sold.
- It is 3 times the initial price if more than 50 % of tickets have been sold.

(c) Construct the method, `updatedTicketPrice()`, that calculates and returns the actual ticket price. This method will be called before a new `Passenger` object is created. [4]

The method `addPassenger()` is called whenever a ticket is purchased by a passenger. This method is passed the name of the passenger and their email address.

This method then updates the `passengers` array and carries out any additional processing.

(d) Construct the method `addPassenger()`. [4]

When the booking process has been completed, the `passengers` array is sorted into ascending order by passenger name. The sort routine used for this is a selection sort named `selectSort()`.

(e) Without writing code, describe the steps carried out by the method `selectSort()`. [5]

(Option A continues on the following page)

(Option A continued)

4. Airport codes are used in place of airport names when displaying flight information. Some examples are shown in **Table 3**.

Table 3: Examples of airport codes

Airport code	Airport name
LHR	London Heathrow
SIN	Singapore
LAX	Los Angeles

(a) Describe how a HashMap could be set up and used to return the airport name when an airport code is passed to the HashMap as a parameter. [3]

An airline stores the email addresses of the passengers in a separate array for each flight. It uses these arrays to send out promotional emails. Many passengers will have booked more than one flight with this airline.

(b) Outline how a HashSet can be used to ensure that only one email is sent out to each email address. [2]

Two HashSets, `set1` and `set2`, have been created:

```
set1: [3,6,2,9,10]
set2: [2,6,1,11]
```

(c) State the output after each of the following coding statements is run:

(i) `System.out.println(set1.contains(1));.` [1]

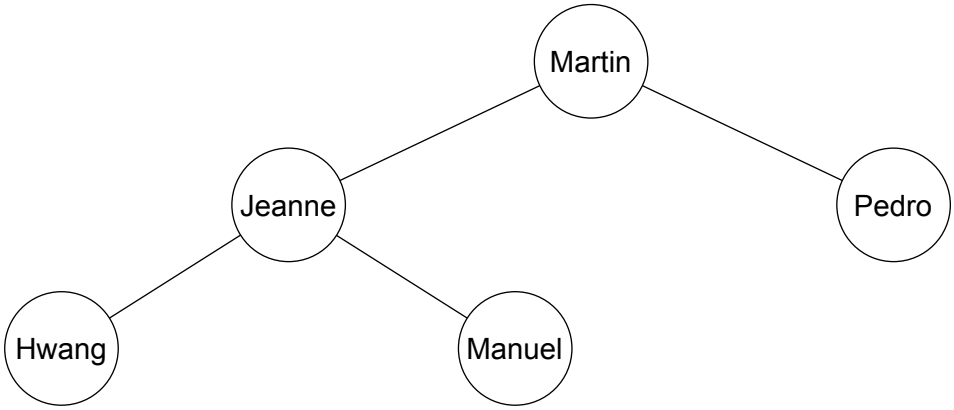
(ii)
```
set1.retainAll(set2);
System.out.println(set1);.
```
[1]

(Option A continues on the following page)

(Option A continued)

5. An array of names is being added to a binary search tree using a recursive insertion method. The tree, after the insertion of the first five names, is shown in **Figure 2**.

Figure 2: The binary search tree after insertion of the first five names



(a) Sketch the binary search tree after the names “Louis” and “Indie” are inserted into it. [2]

(b) Explain why the process of inserting a name into the binary search tree is recursive. [3]

The iterative method `iter(int x)` is shown:

```
public int iter(int x)
{
    s = 0;
    while (x > 0)
    {
        s = s + x;
        x = x -1;
    }
    return s;
}
```

(c) (i) State the value returned by the call `int result = iter(6);`. [1]

(ii) Construct the recursive method `recur(int x)` that performs the same function as the `iter(int x)` method. [4]

(iii) Explain why a run time error might occur in `recur()` for large values of `x`. [3]

(Option A continues on the following page)

(Option A continued)

6. A computer game stores the name and score of each player in the nodes of a linked list.

The linked list makes use of the `Node` class, in which the data for each node consists of the player's name and score. This class is shown:

```
class Node{

    public String name;
    public int score;
    public Node next;

    public Node(String n, int s)
        this.name = n;
        this.score = s;
        this.next = null; // initialize next pointer as null
}
```

The `Node` object `head` will point to the first node in the linked list.

The driver class contains the method `findPlayer()`.

This method is passed the name of a player being searched for in the linked list. It will print the player's score if the name of the player is found, or a suitable message if not found.

(a) Construct the method `findPlayer()`. [5]

The driver class will permanently maintain the nodes in the linked list in descending order of score without ever needing to sort them.

The method `addInOrder()` will accept a new node as a parameter. This node, which contains the player's data, will be inserted into the correct position in the linked list.

(b) Without writing code, describe the steps in the method `addInOrder()`.

Assume the list is not empty. [5]

End of Option A

Option B — Python

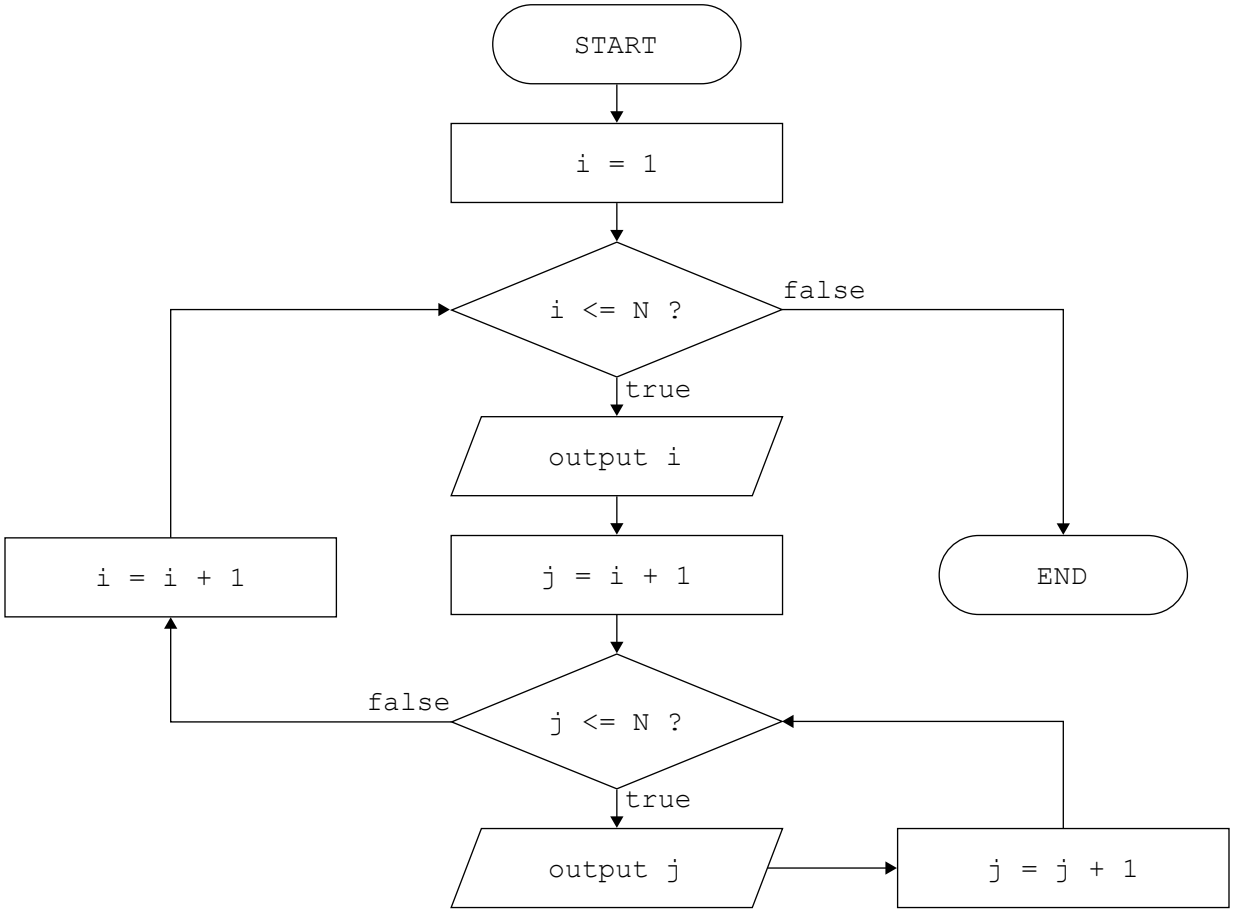
Answer **all** questions. All code that forms part of your responses must be written in Python.

7. Algorithms can be represented by flowcharts.

(a) State **one** reason why a flowchart could be used to represent an algorithm. [1]

Consider the flowchart shown in **Figure 3**.

Figure 3: Flowchart for an algorithm



(b) Copy and complete the trace table formed by the processing of this flowchart, where `N = 3`. [3]

i	i <= N?	j	j <= N ?	output
1				

(Option B continues on the following page)

(Option B, question 7 continued)

Big O notation is used to indicate the time complexity of an algorithm.

(c) State the Big O notation for the algorithm shown in **Figure 3**. [1]

The text file, employees, contains the following information about each employee in a company:

- Surname
- First name
- Role

Each item of data is stored in a new line. Details of the first three employees are shown:

Roberts
John
Sales
Saied
Abdullah
Manager
Sanchez
Pedro
IT manager
…

An algorithm reads the data from the file, employees, line by line. The algorithm writes only the surname of each employee into a new text file, surnames. Both files are currently closed.

(d) Describe the steps that would be followed by this algorithm. [5]

(Option B continues on page 17)

Blank page

(Option B, question 7 continued)

Stacks are a data structure frequently used in computer science to represent real-world scenarios.

Table 4 shows the fundamental operations that operate on a stack.

Table 4: Fundamental operations that operate on a stack

`push()`	Inserts an element onto the top of the stack.
`pop()`	Removes an element from the top of the stack.
`peek()`	Returns the top element of the stack without removing it from the stack.
`isEmpty()`	Returns true if the stack is empty, else it returns false.

(e) Sketch the stack resulting from the following operations.

```
push(1)
push(2)
push(9)
peek()
pop()
push(8)
push(10)
pop()
push(6)
```

You should clearly indicate the top of the stack. [2]

A new stack has been created and filled with elements.

(f) Using only the stack operations shown in **Table 4**, describe the steps that will locate **and** print only the **first** element that was pushed onto the new stack. [3]

(Option B continues on the following page)

(Option B continued)

8. A theatre stores the names and email addresses of its clients in two separate one-dimensional (1D) lists, where each client's name and email address have corresponding indices. The two lists are shown in **Table 5**.

Table 5: Name and email lists

Name list

Index	Name
0	Maria Gonzalez
1	Olga Sarkozky
2	Julia Brown
3	…

Email list

Index	Email address
0	magonzalez24@example.net
1	osarkozky12@one2seven.org
2	jbrown23@example.com
3	…

(a) Explain why lists have been chosen to hold this data instead of separate variables. [3]

A function has been constructed that will change each name so that the surname appears before the first name. For example, "Maria Gonzalez" will become "Gonzalez Maria".

(b) Describe the steps involved in this function.

Assume that there is only one space between each first name and surname. [4]

Consider the following two-dimensional (2D) list, `MAT`.

```
MAT = [[2, 7, 1],
       [8, 9 ,0]]
```

(c) State the code that will identify the data item `0` in `MAT`. [1]

(Option B continues on the following page)

(Option B, question 8 continued)

The following code operates on `MAT`:

```
a = 0
b = 0
for row in range (0, 2):
    for col in range (0, 3):
        a = a + 1
        item = MAT[row][col]
        if item%2 != 0:
            b = b + item
print ('The value of a is : ',a)
print ('The value of b is : ',b)
```

(d) State the output from this code. [2]

The theatre employs a 2D list to indicate the status of each seat for a performance. `True` indicates that the seat has been sold for that performance. `False` indicates that it has not been sold.

The 2D list `seats` is used to represent the seating plan in the theatre. The list contains 40 rows and 30 columns.

For example, `seats[2][3] = False` shows that the seat in row 2, column 3 of the 2D list, `seats`, has not been sold.

After each performance, the sales data is analysed.

One program uses the function `least()`. This accepts the 2D list `seats` and outputs the row for which the lowest number of seats were sold.

(e) Construct the function `least()`.

Assume only one row matches this condition. [5]

(Option B continues on the following page)

(Option B continued)

9. An airline uses object-oriented programming (OOP) to manage its booking system. This includes the class `Passenger`, which is partially shown.

```
class Passenger():

    def __init__(self, name, email, price):
        self._passenger_name = name
        self._email = email
        self._ticket_price = price  # the price is determined when each
                                    # ticket is purchased


    def get_passenger_name(self):
        return self._passenger_name
```

(a) (i) List the operations carried out when the following code is run:

```
p1 = Passenger("Conejo Jorge", "st34@one2seven.org", 240)
```

[3]

The underscore character in `self._email = email` identifies the variable as being a private attribute of the class.

(ii) Outline **two** consequences of declaring an attribute as private. [2]

(Option B continues on the following page)

(Option B, question 9 continued)

The data of each flight are stored in a `Flight` object. This includes details of the passengers stored in a list of `Passenger` objects.

The `Flight` class is partially shown:

```
class Flight():

    def __init__(self, mp, ip):
        self.seats_booked = 0    # incremented every time a ticket is bought
        self.max_passengers = mp # maximum number of passengers for this flight
        self.initial_price = ip  # initial price of a ticket on this flight
        self.passengers = []     # list of Passenger objects

    def updated_ticket_price() # code missing

    def add_passenger() # code missing

    def select_sort() # code missing
```

Multiple classes have a variety of relationships, including aggregation and composition.

(b) Outline the relationship between the `Passenger` and `Flight` classes. [2]

(Option B continues on the following page)

(Option B, question 9 continued)

The initial price of a ticket for each flight (`initial_price`) is the value set by the airline before any tickets have been sold for that flight. However, the actual ticket price that the passenger pays (`ticket_price`) will change depending on the number of tickets already sold.

The airline determines the actual price of a ticket as follows:
- It is equal to the initial price if less than 10 % of tickets have been sold.
- It is 1.5 times the initial price if 10 %–50 % of tickets have been sold.
- It is 3 times the initial price if more than 50 % of tickets have been sold.

(c) Construct the function, `updated_ticket_price()`, that calculates and returns the actual ticket price. This function will be called before a new `Passenger` object is created. [4]

The method `add_passenger()` is called whenever a ticket is purchased by a passenger. This method is passed the name of the passenger and their email address.

This method then updates the `passengers` list and carries out any additional processing.

(d) Construct the method `add_passenger()`. [4]

When the booking process has been completed, the `passengers` list is sorted into ascending order by passenger name. The sort routine used for this is a selection sort named `select_sort()`.

(e) Without writing code, describe the steps carried out by the method `select_sort()`. [5]

(Option B continues on the following page)

(Option B continued)

10. Airport codes are used in place of airport names when displaying flight information. Some examples are shown in **Table 6**.

Table 6: Examples of airport codes

Airport code	Airport name
LHR	London Heathrow
SIN	Singapore
LAX	Los Angeles

(a) Describe how a dictionary could be set up and used to return the airport name when an airport code is passed to the dictionary as a parameter. [3]

An airline stores the email addresses of the passengers in a separate list for each flight. It uses these lists to send out promotional emails. Many passengers will have booked more than one flight with this airline.

(b) Outline how a set can be used to ensure that only one email is sent out to each email address. [2]

Two sets, `set1` and `set2`, have been created:

```
set1 = {3,6,2,9,10}
set2 = {2,6,1,11}
```

(c) State the output after each of the following coding statements is run:

(i) `print(12 in set1).` [1]

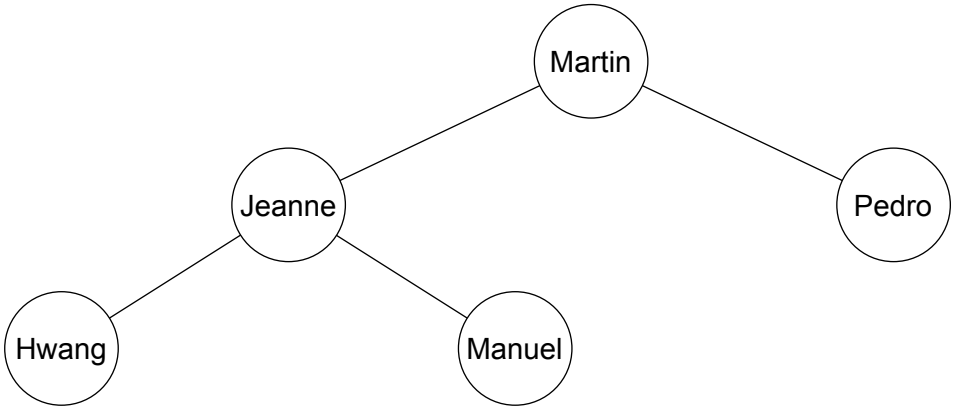
(ii) `print(set1 & set2).` [1]

(Option B continues on the following page)

(Option B continued)

11. A list of names is being added to a binary search tree using a recursive insertion method. The tree, after the insertion of the first five names, is shown in **Figure 4**.

Figure 4: The binary search tree after insertion of the first five names



(a) Sketch the binary search tree after the names “Louis” and “Indie” are inserted into it. [2]

(b) Explain why the process of inserting a name into the binary search tree is recursive. [3]

The iterative method `iter(x)` is shown:

```
def iter(x):
    s = 0
    while (x > 0):
        s = s + x
        x = x -1
    return s
```

(c) (i) State the value returned by the call `result = iter(6)`. [1]

(ii) Construct the recursive method `recur(x)` that performs the same function as the `iter(x)` method. [4]

(iii) Explain why a run time error might occur in `recur()` for large values of `x`. [3]

(Option B continues on the following page)

(Option B continued)

12. A computer game stores the name and score of each player in the nodes of a linked list.

The linked list makes use of the `Node` class, in which the data for each node consists of the player's name and score. This class is shown:

```
class Node():

    def __init__(self, name, score):
        self.name = name
        self.score = score
        self.next = None    # initialize next pointer as null
```

The `Node` object `self.head` will point to the first node in the linked list.

The driver code contains the method `def find_player(self, name).`

This method is passed the name of a player being searched for in the linked list. It will print the player's score if the name of the player is found, or a suitable message if not found.

(a) Construct the method `def find_player(self, name).` [5]

The driver code will permanently maintain the nodes in the linked list in descending order of score without ever needing to sort them.

The method `def add_in_order(self, new_node)` will accept a new node as a parameter. This node, which contains the player's data, will be inserted into the correct position in the linked list.

(b) Without writing code, describe the steps in the method `def add_in_order(self, new_node):`.

Assume the list is not empty. [5]

End of Option B

Diploma Programme
Programme du diplôme
Programa del Diploma

Markscheme

Specimen paper

Computer science

Higher level

Paper 2

22 pages

International Baccalaureate®
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General marking instructions

1. Follow the markscheme provided, award only whole marks and mark only in **RED**.

2. Make sure that the question you are about to mark is highlighted in the mark panel on the right-hand side of the screen.

3. Where a mark is awarded, a tick/check (✓) **must** be placed in the text at the **precise point** where it becomes clear that the candidate deserves the mark. **One tick to be shown for each mark awarded**.

4. Sometimes, careful consideration is required to decide whether or not to award a mark. In these cases use RM™ Assessor annotations to support your decision. You are encouraged to write comments where it helps clarity, especially for re-marking purposes. Use a text box for these additional comments. It should be remembered that the script may be returned to the candidate.

5. Personal codes/notations are unacceptable.

6. Where an answer to a part question is worth no marks but the candidate has attempted the part question, enter a zero in the mark panel on the right-hand side of the screen. Where an answer to a part question is worth no marks because the candidate has not attempted the part question, enter an “NR” in the mark panel on the right-hand side of the screen.

7. Please ensure you check all scanned pages. The candidate may have answered more than one option.

8. **If a candidate has attempted more than one Option** within a paper, mark all the candidate’s work. RM™ Assessor will only award the marks for the higher scoring Option. Once all the work the candidate has attempted has been marked, please click “COMPLETE”; all the other questions from the other Options will auto complete to “NR” for “no response”.

9. Ensure that you have viewed **every** page including any additional sheets. Please ensure that you stamp “SEEN” on any page that contains no other annotation.

10. A mark should not be awarded where there is contradiction within an answer. Make a comment to this effect using a text box or the “CON” stamp.

Subject details: Computer Science HL Paper 2 markscheme

Mark allocation

Candidates are required to answer **all** questions in **one** Option. Total 80 marks.

General

A markscheme often has more specific points worthy of a mark than the total allows. This is intentional. Do not award more than the maximum marks allowed for that part of a question.

When deciding upon alternative answers by candidates to those given in the markscheme, consider the following points:

- Each statement worth one point has a separate line and the end is signified by means of a semi-colon (;).
- An alternative answer or wording is indicated in the markscheme by a "/"; either wording can be accepted.
- Words in (…) in the markscheme are not necessary to gain the mark.
- If the candidate's answer has the same meaning or can be clearly interpreted as being the same as that in the markscheme then award the mark.
- Mark positively. Give candidates credit for what they have achieved and for what they have got correct, rather than penalizing them for what they have not achieved or what they have got wrong.
- Remember that many candidates are writing in a second language; be forgiving of minor linguistic slips. In this subject effective communication is more important than grammatical accuracy.
- Occasionally, a part of a question may require a calculation whose answer is required for subsequent parts. If an error is made in the first part then it should be penalized. However, if the incorrect answer is used correctly in subsequent parts then **follow through** marks should be awarded. Indicate this with "**FT**".

General guidance

Issue	Guidance
Answering more than the quantity of responses prescribed in the questions	• In the case of an "identify" question, read all answers and mark positively up to the maximum marks. Disregard incorrect answers. • In the case of a "describe" question, which asks for a certain number of facts *eg* "describe two kinds", mark the first two correct answers. This could include two descriptions, one description and one identification, or two identifications. • In the case of an "explain" question, which asks for a specified number of explanations *eg* "explain two reasons …", mark the first two correct answers. This could include two full explanations, one explanation, one partial explanation *etc*.

Option A — Java

1. (a) *Award **[1 max]***

because it is in diagrammatic form;
it is program agnostic;
no complicated code / syntax to follow;

(b) *Award **[3]***

correct values of i;
correct values of j;
correct output;

i	i < = N ?	j	j < = N ?	output
1	TRUE			1
		2	TRUE	2
		3	TRUE	3
		4	FALSE	
2	TRUE			2
		3	TRUE	3
		4	FALSE	
3	TRUE	4	FALSE	3
4	FALSE			

(c) *Award **[1]**. Note: the algorithm shows a nested loop.*

On^2;

(d) *Award **[5 max]** as follows*

opening *employees* file for reading / default mode;
open *surnames* file for writing;
looping through *employees* file;
use of a counter or equivalent;
writing surnames only to *surnames* file;
closing both files;

*Note: Award **[1]** for first 2 marking points if files are opened without indicating access modes.*

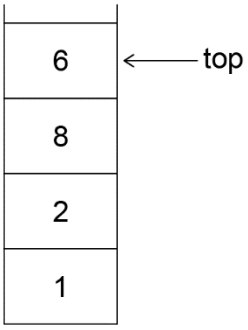
Algorithm is:

```
open employees file for reading / in default mode
open surnames file for writing
set counter to 0
repeat until end of employees file
    read next line from employees file
    if counter%3 = 0 *
        write line to surnames file
    increment counter
close both files
```

* there are alternatives ways of using counter

(e) *Award **[2]**.*

correct elements in stack **[1]**;
correct indication of top **[1]**;



(f) *Award **[3]** as follows:*

*Award **[1]** for each correct line (or equivalent).*

repeat until `isEmpty` is `true` / while `isEmpty` is `false`;
pop an element and save its value/save in a variable/push it onto a new stack (stack2)
print the last value saved/print the variable/pop stack2 and print value

Note: Answer must demonstrate an algorithmic approach.

2. (a) *Award **[3 max]***

It is easier to work with one single structure than several individual ones;
Individual elements that have the same value can be quickly accessed via their indices;
Iteration through arrays is more efficient / Methods associated with arrays can make operations more efficient;

(b) *Award **[4 max]***

repeat for each entry/index in the `Name` array
    identify the position of the space in each name
    save the substring to the left as s1 and the substring to the right as s2
    save the new entry as s2+space+s1

(c) *Award **[1]***

```
MAT[1][2];
```

(d) *Award **[1]** each;*

The value of a is: 6
The value of b is: 17

(e) *Award **[5 max]** as follows:*

*Award **[1]** for initial variables correctly set;*
*Award **[1]** for both loops;*
*Award **[1]** for checking seat for* `False`*;*
*Award **[1]** for updating row value;*
*Award **[1]** for resetting* `tempTotal`*;*
*Award **[1]** for printing the correct row;*

```
public void least(boolean[][]seats)
{
    int falseTotal = 0;
    int leastRow = 0;

    for (int x = 0; x < 40; x++)
    {
        int tempTotal = 0;
        for (int y = 0; y < 30; y++)
        {
         if (seats[x][y] == false)       // if(!seats[x][y] )
            tempTotal = tempTotal + 1;
        }
        if (tempTotal > falseTotal)
        {
            leastRow = x;
            falseTotal = tempTotal;
        }
    }
    System.out.println ("Row with least seats sold is " + leastRow);
}
```

3. (a) (i) *Award **[3 max]***

The `(Passenger)` constructor is called;
A new object is created / memory is assigned for a new object ...;
… identified by the variable `p1`;
The `passengerName`, `email` and `ticketPrice` variables / all of the variables are initialized;

(ii) *Award **[2 max]***

provides greater security for the values stored in these variables/for the code;
restricts access to such variables <u>from outside of the class</u> /only allows access through public / getter, setter methods;
(helps to) prevents <u>accidental/unintended</u> changes to the variables (from outside of the class);

(b) *Award **[2 max];***

composition;
The variable `ticketPrice` in the `Passenger` class depends upon the `Flight` class;

(c) *Award **[4 max]***

*Award **[1]** for calculating the % seats sold;*
*Award **[1]** for all 3 conditions correct;*
*Award **[1]** for if … else structure correct;*
*Award **[1]** for return;*

```
public double updatedTicketPrice()
{
    double price;
    double percentSold = (seatsBooked*100)/maxPassengers;
    if (percentSold >= 10 && percentSold <= 50)
        price = initialPrice*1.5;
    else
        if (percentSold > 50)
            price = initialPrice*3;
        else
            price = initialPrice;
    return price;
}
```

(d) *Award **[4 max]***

*Award **[1]** for calling* `updatedTicketPrice();`
*Award **[1]** for creating new* `Passenger` *object*;
*Award **[1]** for updating* `passengers` *array;*
*Award **[1]** for incrementing* `seatsBooked`;

```
public void addPassenger(String name, String email)
{
    double s = updatedTicketPrice();
    Passenger p = new Passenger(name,email,s);
    passengers[seatsBooked] = p;
    seatsBooked = seatsBooked + 1;
}
```

(e) *Award **[5 max]***

the name field of each `Passenger` object is used in the sorting process;
the array is divided into two parts: a sorted subset and an unsorted subset;
the smallest element is found in the unsorted part of the array and compared to the first (leftmost) element in the unsorted set;
if smaller, the two objects are swapped (and the first element in the unsorted set joins the sorted set);
the two steps above are repeated until the whole array is sorted / there are no more elements in the unsorted set;

4. (a) *Award **[3 max]***

The *HashMap* is set up using pairs of string values;
representing the airport code and airport name / diagram to show this;
when the airport code is used as a parameter for the *HashMap,* the matching/corresponding value of the airport name will be returned;

(b) *Award **[2 max]***

the arrays are combined to form a single set;
the set will automatically eliminate any duplicates / will only hold unique values;

(c) (i) *Award **[1 max]***

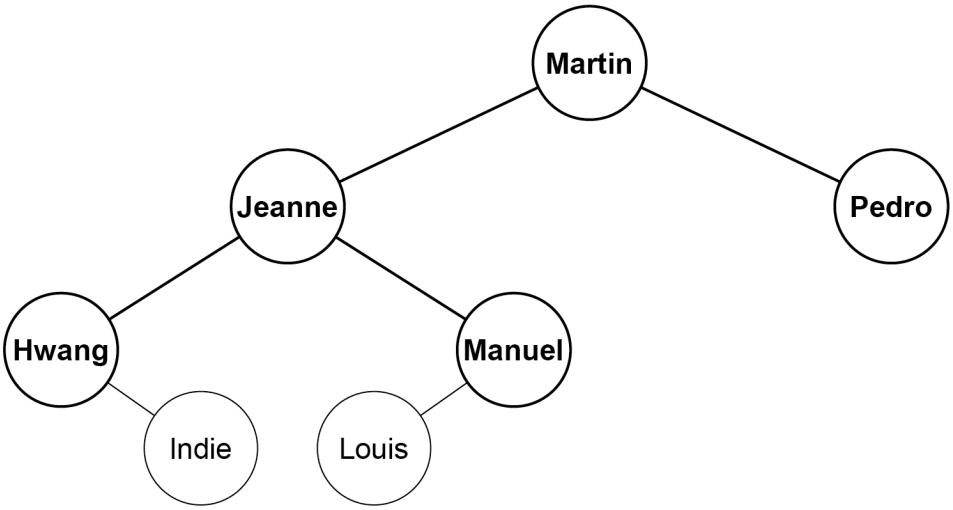
false;

(ii) *Award **[1 max]***

[2,6];

5. (a) *Award **[2 max]***

*award **[1]** for each correct entry;*



(b) *Award **[3 max]***

The insert method is initially passed the root of the tree as a parameter (where a comparison is made);
Then either the left or right subtree is selected (depending upon the result of the comparison);
The method is called again, using the root of this subtree as the parameter;
This is repeated until the insertion point is found (base case) / subtree is null;

(c) (i) *Award **[1 max]***

21;

(ii) *Award **[4 max]***

*Award **[1]** for including a recursive call that will lead to the base case;*
*Award **[1]** for having a base case;*
*Award **[1]** for return + int;*
*Award **[1]** for a completely correct algorithm;*

```
public int recur(int x)
{
    if (x != 0)
        return x + recur(x-1);
    else
        return x;
}
```

(iii) *Award **[3 max]***

data is stored in the memory / on a stack <u>every time a recursive call is made</u>;
if the value of `x` is very large ...;
… the stack may overflow / system may run out of memory;

6. (a) *Award **[5 max]***

*Award **[1]** for making use of a temporary node;*
*Award **[1]** for Boolean;*
*Award **[1]** for list traversal;*
*Award **[1]** for ending traversal if found;*
*Award **[1]** for correct comparison;*
*Award **[1]** for returning the correct values (in both cases);*

```
public void findPlayer(String name)
{
    Node temp = head;
    boolean found  = false;
    while(temp != null)
    {
        if (temp.name.compareTo(name) == 0)  // allow = name
        {
            System.out.println(temp.score);
            found = true;
            break;
        }
        else
            temp = temp.next;
    }
    if (found  == false)
        System.out.println("Not Found");
}
```

(b) *Award **[5 max]***

The algorithm will:

- make a temporary node equal to `head`;
- make use of a "previous" node;
- traverse the linked list (using the temporary node) comparing the score field of each node to the score field of the new node;
- until the insertion point is found;
- if insertion is at the beginning, link the new node to the head and make the head equal to the new node;
- if insertion is at the end, link the previous node to the new node;
- copy the *next* pointer of the node previous to the insertion point to the *next* pointer of the new node;
- link the previous node to the new node;

Note: Allow the use of diagrams to illustrate any of the above points.

Option B — Python

7. (a) *Award **[1 max]***

because it is in diagrammatic form;
it is program agnostic;
no complicated code / syntax to follow;

(b) *Award **[3]***

correct values of i;
correct values j;
correct output;

`i`	`i < = N ?`	`j`	`j < = N ?`	`output`
`1`	TRUE			1
		2	TRUE	2
		3	TRUE	3
		4	FALSE	
2	TRUE			2
		3	TRUE	3
		4	FALSE	
3	TRUE	4	FALSE	3
4	FALSE			

(c) *Award **[1]**. Note: the algorithm shows a nested loop.*

On^2;

(d) *Award **[5 max]** as follows*

opening *employees* file for reading / default mode;
open *surnames* file for writing;
looping through *employees* file;
use of a counter or equivalent;
writing surnames only to *surnames* file;
closing both files;

*Note: Award **[1]** for first 2 marking points if files are opened without indicating access modes*

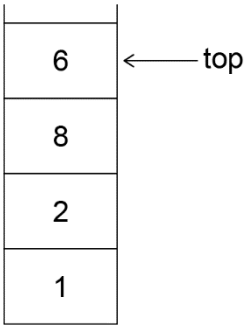
Algorithm is:

```
open employees file for reading / in default mode
open surnames file for writing
set counter to 0
repeat until end of employees file
    read next line from employees file
    if counter%3 = 0 *
        write line to surnames file
    increment counter
close both files
```

* there are alternatives ways of using counter

(e) *Award **[2]**.*

correct elements in stack **[1]**;
correct indication of top **[1]**;



(f) *Award **[3]** as follows:*

*Award **[1]** for each correct line (or equivalent).*

repeat until `isEmpty` is `true` / while `isEmpty` is `false`;
    pop an element and save its value/save in a variable/push it onto a new stack (stack2);
print the last value saved/print the variable/pop stack2 and print value;

Note: Answer must demonstrate an algorithmic approach.

8. (a) *Award **[3 max]***

It is easier to work with one single structure than several individual ones;
Individual elements that have the same value can be quickly accessed via their indices;
Iteration through arrays is more efficient / Methods associated with arrays can make operations more efficient;

(b) *Award **[4 max]** as follows:*

repeat for each entry/index in the `Name` list
identify the position of the space in each name
save the substring to the left as s1 and the substring to the right as s2
save the new entry as s2+space+s1

repeat for each entry/index in the `Name` list
use the `split()` function
to separate the 2 strings into 2 list values (s1 and s2)
save the new entry as s2+space+s1

(c) *Award **[1]***

```
MAT[1][2];
```

(d) *Award **[1]** for each:*

The value of a is: 6
The value of b is: 17

(e) *Award **[5 max]** as follows:*

*Award **[1]** for initial variables correctly set;*
*Award **[1]** for both loops;*
*Award **[1]** for checking seat for* `False`*;*
*Award **[1]** for updating row value;*
*Award **[1]** for resetting* `temp_total`*;*
*Award **[1]** for printing the correct row;*

```
def least(seats):

    false_total = 0
    least_row = 0

    for x in range(40):
        temp_total = 0
        for y in range (30):
            if seats[x][y] == False:
                temp_total = temp_total + 1
        if temp_total > false_total:
            least_row = x
            false_total = temp_total

    print ("Row with least seats sold is ",least_row)
```

9. (a) (i) *Award **[3 max]***

The `(Passenger)` constructor is called;
A new object is created / Memory is assigned for a new object ...;
… identified by the variable `p1`;
The `passenger_name`, `email` and `ticket_price` variables / all of the variables are initialized;

(ii) *Award **[2 max]***

provides greater security for the values stored in these variables/for the code;
restricts access to such variables from outside of the class /only allows access through public / getter, setter methods;
(helps to) prevents accidental/unintended changes to the variables (from outside of the class);

(b) *Award **[2 max]***

composition;
The variable `ticket_price` in the `Passenger` class depends upon the `Flight` class;

(c) *Award **[4 max]***

*Award **[1]** for calculating the % seats sold;*
(allow `len(self.passengers)` for `self.seats_booked`)
*Award **[1]** for all 3 conditions correct;*
*Award **[1]** for if … else structure correct;*
*Award **[1]** for return;*

```
def updated_ticket_price(self):
    percent_sold = self.seats_booked * 100 / self.max_passengers
    if percent_sold >= 10 and percent_sold <= 50.0:
        price = self.initial_price * 1.5
    elif percent_sold > 50.0:
        price =  self.initial_price * 3
    else:
        price = self.initial_price
    return price
```

(d) *Award **[4 max]***

*Award **[1]** for calling* calling `updated_ticket_price()`;
*Award **[1]** for creating new* `Passenger` object;
*Award **[1]** for updating* `passengers` list;
*Award **[1]** for incrementing* `seats_booked`;

```
def add_passenger(self, name, email):
    s = self.updated_ticket_price()
    p = Passenger(name, email, s)
    self.passengers.append(p)
    self.seats_booked = self.seats_booked + 1
```

(e) *Award **[5 max]***

the name field of each `Passenger` object is used in the sorting process;
the list is divided into two parts: a sorted subset and an unsorted subset;
the smallest element is found in the unsorted part of the list and compared to the first (leftmost) element in the unsorted set;
if smaller, the 2 objects are swapped (and the first element in the unsorted set joins the sorted set);
the two steps above are repeated until the whole list is sorted / there are no more elements in the unsorted set;

10. (a) *Award **[3 max]***

The dictionary is set up using pairs of string values;
representing the airport code and airport name / diagram to show this;
when the airport code is used as a parameter for the dictionary, the matching/corresponding value of the airport name will be returned;

(b) *Award **[2 max]***

the lists are combined to form a single set/HashSet;
the set/HashSet will automatically eliminate any duplicates / will only hold unique values;

(c) (i) *Award **[1 max]***

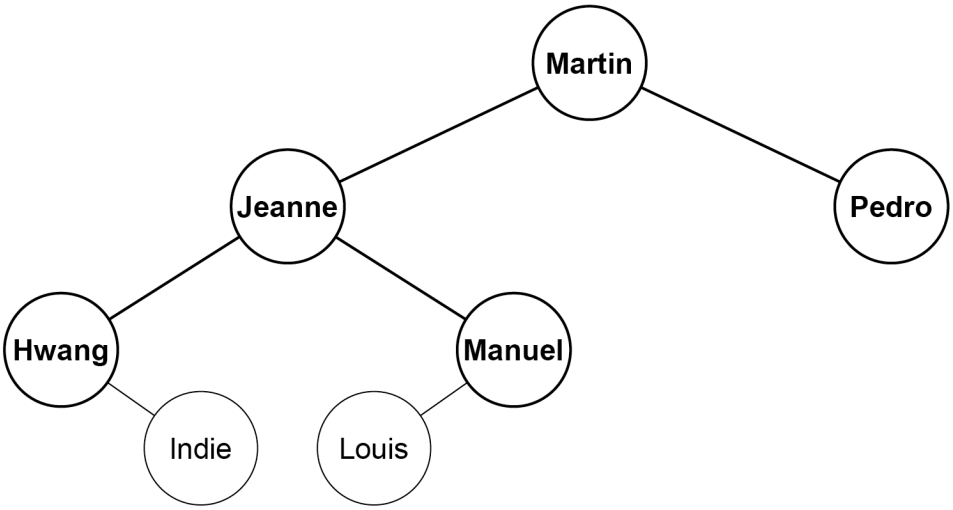
false;

(ii) *Award **[1 max]***

{2, 6};

11. (a) *Award **[2 max]***

*award **[1]** for each correct entry;*



(b) *Award **[3 max]***

The insert method is initially passed the root of the tree as a parameter (where a comparison is made);
Then either the left or right subtree is selected (depending upon the result of the comparison);
The method is called again, using the root of this subtree as the parameter;
This is repeated until the insertion point is found (base case) / subtree is null;

(c) (i) *Award **[1 max]***

21;

(ii) *Award **[4 max]***

*Award **[1]** for including a recursive call that will lead to the base case;*
*Award **[1]** for having a base case;*
*Award **[1]** for return;*
*Award **[1]** for a completely correct algorithm;*

```
def recur(x):
    if x != 0:
        return x + recur(x-1);
    else:
        return x
```

(iii) *Award **[3 max]***

data is stored each in the memory / on a stack every time a recursive call is made;
if the value of `x` is very large ...;
… the stack might overflow / system might run out of memory;

12. (a) *Award **[5 max]***

*Award **[1]** for making use of a temporary node;*
*Award **[1]** for Boolean;*
*Award **[1]** for list traversal;*
*Award **[1]** for ending traversal if found;*
*Award **[1]** for correct comparison;*
*Award **[1]** for returning the correct values (in both cases);*

```
def find_player(self, name):
    temp = self.head
    found  = False
    while temp != None:
        if temp.name == name:
            found = True
            print (temp.score)
            break
    else:
            temp = temp.next
    if found == False:
        print ("Not found")
```

(b) *Award **[5 max]***

The algorithm will:

- make a temporary node equal to `head`;
- make use of a "previous" node;
- traverse the linked list (using the temporary node) comparing the score field of each node to the score field of the new node;
- until the insertion point is found;
- if insertion is at the beginning, link the new node to the head and make the head equal to the new node;
- if insertion is at the end, link the previous node to the new node;
- copy the *next* pointer of the node previous to the insertion point to the *next* pointer of the new node;
- link the previous node to the new node;

Note: Allow the use of diagrams to illustrate any of the above points

Diploma Programme
Programme du diplôme
Programa del Diploma

Computer science
Standard level
Paper 1

Specimen paper

Candidate session number

1 hour 15 minutes

Instructions for candidates

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- A clean copy of the **computer science case study** is required for this examination paper.
- Read the case study carefully.
- Section A: answer all the questions.
- Section B: answer the question.
- Answers must be written within the answer boxes provided.
- The maximum mark for this examination paper is **[50 marks]**.

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Section A

Answer **all** questions from this section. Answers must be written within the answer boxes provided.

1. A hospital has an IT team that repairs computer hardware and software.

(a) (i) Identify the bus that transmits the program counter (PC) contents to the memory address register (MAR). [1]

. . . . . .
. . . . . .

(ii) Outline the function of an accumulator. [2]

. . . . . .
. . . . . .
. . . . . .
. . . . . .

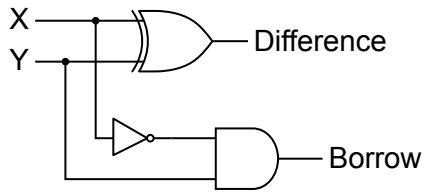
(b) State the binary equivalent of the hexadecimal number AC. [1]

. . . . . .
. . . . . .

(This question continues on the following page)

(Question 1 continued)

(c) Construct a truth table for the following circuit: [2]



X	Y	Difference	Borrow

The IT team is considering replacing its laptops with ones that use flash memory instead of read-only memory (ROM) for permanent storage of the basic input/output system (BIOS).

(d) Compare and contrast the use of a read-only memory (ROM) chip with flash memory for storing the basic input/output system (BIOS). [3]

. . . . . .
. . . . . .
. . . . . .
. . . . . .
. . . . . .
. . . . . .

(e) Outline the function of queues in multilevel queue scheduling. [2]

. . . . . .
. . . . . .
. . . . . .
. . . . . .

2. A hospital has a local area network (LAN) with a router that uses a fiber-optic cable to connect directly to an internet service provider's equipment.

(a) List **three** advantages of using a fiber-optic cable instead of a twisted-pair cable. [3]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(b) Describe the process of packet switching. [4]

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(This question continues on the following page)

(Question 2 continued)

The hospital's local area network (LAN) is divided into multiple smaller subnetworks, also called subnets.

(c) Explain how subnetting a local area network (LAN) can reduce network congestion. [4]

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3. A hospital stores patients' data in a relational database.

(a) Outline **one** benefit of using a relational database. [2]

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(b) Identify **two** scalability issues in a relational database. [2]

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The database view, VISITS, is an extract showing the dates patients visited the hospital for an appointment.

The VisitDate field represents the date of each patient's appointment.

PatID	PatientName	PatientDOB	DocID	Doctor	VisitDate	Treatment
1	Samuel Moody	2007/06/11	1	Jill Smith	2025/01/11	Bone scan
1	Samuel Moody	2007/06/11	2	Norayr Lomidze	2025/01/12	Physical therapy
2	Lin Kobayashi	1984/02/24	1	Jill Smith	2025/01/12	Physical therapy
3	Yeojoo Kim	1999/08/18	3	Phoebe Drakos	2025/01/13	Radiation therapy
3	Yeojoo Kim	1999/08/18	3	Phoebe Drakos	2025/01/14	Therapy part 1

(This question continues on the following page)

(Question 3 continued)

(c) Construct an SQL SELECT query to list the names of patients whose treatment includes the word “therapy”.

A patient’s name should only appear once. [3]

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The database view, VISITS, can be represented by the following notation:

```
VISITS (PatID, PatientName, PatientDOB, DocID, Doctor,
VisitDate, Treatment)
```

(d) Construct the database in third normal form (3NF) for the three entities in the VISITS view extract. [5]

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4. A hospital uses a supervised learning algorithm for diagnosis.

(a) Identify **two** factors that might determine the need for an increase in processing power for this supervised learning algorithm. [2]

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An article about algorithmic bias in machine learning appeared in a national newspaper. Some of the hospital's patients became concerned about the ethics of using supervised learning for diagnosis.

(b) Outline **one** strategy the hospital could use to address patients' concerns. [2]

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Section B

Answer the following question, which refers to the case study. Answers must be written within the answer boxes provided.

5. Ransomware and rootkit are types of malware that result in system downtime and data losses.

(a) Outline **one other** potential consequence of a rootkit attack on a hospital. [2]

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(b) Outline **two** ways in which a firewall protects a network system from cybersecurity attacks. [4]

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(This question continues on the following page)

(Question 5 continued)

Jane Wang, Head of Cybersecurity at TaylorMade Healthcare, highlighted the importance of being proactive and creating a culture of security awareness.

(c) To what extent do the benefits of implementing a bring your own device (BYOD) policy outweigh the risks associated with rootkit and ransomware attacks? [6]

Please **do not** write on this page.

Answers written on this page
will not be marked.

Diploma Programme
Programme du diplôme
Programa del Diploma

Markscheme

Specimen paper

Computer science

Standard level

Paper 1

13 pages

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General marking instructions

1. Follow the markscheme provided, award only whole marks and mark only in **RED**.

2. Make sure that the question you are about to mark is highlighted in the mark panel on the right-hand side of the screen.

3. Where a mark is awarded, a tick/check (✓) **must** be placed in the text at the **precise point** where it becomes clear that the candidate deserves the mark. **One tick to be shown for each mark awarded**.

4. Sometimes, careful consideration is required to decide whether or not to award a mark. In these cases use RM™ Assessor annotations to support your decision. You are encouraged to write comments where it helps clarity, especially for re-marking purposes. Use a text box for these additional comments. It should be remembered that the script may be returned to the candidate.

5. Personal codes/notations are unacceptable.

6. Where an answer to a part question is worth no marks but the candidate has attempted the part question, enter a zero in the mark panel on the right-hand side of the screen. Where an answer to a part question is worth no marks because the candidate has not attempted the part question, enter an "NR" in the mark panel on the right-hand side of the screen.

7. Please ensure you check all scanned pages. The candidate may have answered more than one option.

8. **If a candidate has attempted more than one Option** within a paper, mark all the candidate's work. RM™ Assessor will only award the marks for the higher scoring Option. Once all the work the candidate has attempted has been marked, please click "COMPLETE"; all the other questions from the other Options will auto complete to "NR" for "no response".

9. Ensure that you have viewed **every** page including any additional sheets. Please ensure that you stamp "SEEN" on any page that contains no other annotation.

10. A mark should not be awarded where there is contradiction within an answer. Make a comment to this effect using a text box or the "CON" stamp.

Subject details: Computer Science SL Paper 1 markscheme

Mark allocation

Section A: Candidates are required to answer **all** questions. Total 38 marks.
Section B: Candidates are required to answer **all** questions. Total 12 marks.
Maximum total = 50 marks.

General

A markscheme often has more specific points worthy of a mark than the total allows. This is intentional. Do not award more than the maximum marks allowed for that part of a question.

When deciding upon alternative answers by candidates to those given in the markscheme, consider the following points:

- Each statement worth one point has a separate line and the end is signified by means of a semi-colon (;).
- An alternative answer or wording is indicated in the markscheme by a "/"; either wording can be accepted.
- Words in (…) in the markscheme are not necessary to gain the mark.
- If the candidate's answer has the same meaning or can be clearly interpreted as being the same as that in the markscheme then award the mark.
- Mark positively. Give candidates credit for what they have achieved and for what they have got correct, rather than penalizing them for what they have not achieved or what they have got wrong.
- Remember that many candidates are writing in a second language; be forgiving of minor linguistic slips. In this subject effective communication is more important than grammatical accuracy.
- Occasionally, a part of a question may require a calculation whose answer is required for subsequent parts. If an error is made in the first part then it should be penalized. However, if the incorrect answer is used correctly in subsequent parts then **follow through** marks should be awarded. Indicate this with "**FT**".

General guidance

Issue	Guidance
Answering more than the quantity of responses prescribed in the questions	• In the case of an "identify" question, read all answers and mark positively up to the maximum marks. Disregard incorrect answers. • In the case of a "describe" question, which asks for a certain number of facts *eg* "describe two kinds", mark the first two correct answers. This could include two descriptions, one description and one identification, or two identifications. • In the case of an "explain" question, which asks for a specified number of explanations *eg* "explain two reasons …", mark the first two correct answers. This could include two full explanations, one explanation, one partial explanation *etc*.

Section A

1. (a) (i) *Award **[1]** max*

Address (bus).

(ii) *Award **[2]** max*

To store intermediate arithmetic and logic results/hold operand data;
…to reduce the number of memory accesses (by holding intermediate results);

Acts as a temporary storage location;
…which stores the values and results from arithmetic and logical operations;

Holds data or the results of an operation;
…during the processing cycles;

(b) *Award **[1]** max*

10101100

(c) *Award **[2]** max*

*Award **[1]** for each correct column (Difference, Borrower)*

X	Y	Difference	Borrower
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

(d) *Award **[3]** max*

*Award up to **[1]** per difference*

(Permanence) ROM is read-only and non-volatile, whereas flash data can be electronically erased and reprogrammed;
(Security) ROM is more secure (less corruptible) than flash due to its non-volatility;
(Longevity) ROM doesn't wear from read/write whereas flash has a limited lifespan;
(Size/capacity) Flash could be larger capacity than ROM;
(Update) Flash can be updated for user preferences whereas ROM needs CMOS RAM to store.

(e) *Award **[2]** max*

Award [1] for definition, and [1] for expansion

Processes are divided into different queues;
…based on specific criteria, e.g., separate queues for system processes, interactive user processes, background processes;

Processes are divided into different queues;
…(to allow the operating system) to allocate resources more efficiently;

Processes are divided into different queues;
…to ensure higher priority processes receive more immediate attention;

Processes are divided into different queues;
…to balance the load across different types of processes/ensure no single process monopolizes the CPU;

2. (a) *Award **[3]** max*

*Award **[1]** per point*

Transmission speed/higher bandwidth;
Lower latency compared to twisted-pair cables;
Transmits signals over greater distances than twisted-pair cables;
Immune to electromagnetic interference (EMI) and radio-frequency interference (RFI);
Better security against unauthorized access or data tapping;
Better scalability as data transmission needs grow;
Generally more durable than twisted-pair cables;
Longer lifespan and lower maintenance requirements;

(b) *Award **[4]** max*

*Award **[1]** per line*

The original data is divided into smaller units called packets;
Each packet is labeled (header/source and destination IP addresses, etc.);
Packets are sent over the network individually, taking different routes;
Error checking is performed at each switch/router;
At destination, they are reassembled using sequence numbers in the packet headers;
The recipient system acknowledges receipt/requests missing packets to be resent.

(c) *Award **[4]** max*

Award [1] for identifying a point and [1] for expansion for any two points

Broadcasts: Traffic is confined to smaller subnets/few devices on the subnet;
Devices receive fewer broadcast messages freeing up bandwidth;

Specialist transmissions: By subnetting segments traffic based on departments, the type of data traffic is separated out, avoiding data collisions;

Network stability: Less competition for resources;
To ensure the network is more stable (with fewer problems to affect congestion);

Isolation of problems: Packet collisions and broadcast storms can be isolated within a subnet, preventing them from affecting the entire network;

Prioritizing: QoS strategies to prioritize certain types of traffic;
Ensuring critical transmissions are unaffected by less critical ones.

Optimized performance: Reduced congestion;
Allowing the network to perform at an optimized level.

3. (a) *Award **[2]** max*

*Award **[1]** for stating a benefit and **[1]** for expansion*

Provides data integrity;
Accurate and reliable data/primary and foreign keys prevent orphan data/enforce relationships;

Maintains data consistency;
Ensures data in one part of the database does not contradict data in another;

Allows data concurrency;
Multiple users can access data at the same time;

Reduces of data redundancy;
Uses normalization to ensure data is only stored once;

Allows for complex querying;
Uses a declarative/industry standard language;

(b) *Award **[1]** per point, up to **[2]** max*

Data volume growth (leading to performance issues);
Increased concurrency (an increase in users may lead to concurrency issues);
Complexity of queries (more sophisticated analysis use more resources);
Hardware limitations (single server becomes overloaded);
Load balancing on multiple servers;
Inefficient indexing (affecting time to search);

(c) *Award **[3]** max*

*Award **[1]** for SELECT DISTINCT or GROUP BY*

*Award **[1]** for using LIKE*

*Award **[1]** for using the wildcard (any wildcard accepted) and therapy*

```
SELECT DISTINCT PatientName FROM VISITS
WHERE Treatment LIKE '%therapy%';
```

Alternative solution using group by:

```
SELECT PatientName FROM VISITS
WHERE Treatment LIKE '%therapy%'
GROUP BY PatientName
```

Alternative solution using a subquery and group by:

```
SELECT PatientName
FROM (
    SELECT PatientName
    FROM VISITS
    WHERE Treatment LIKE '%therapy%'
) AS subquery
GROUP BY PatientName;
```

(d) *Award **[5]** max*

*Award **[1]** for the PATIENT table with correct Primary Key*

*Award **[1]** for the DOCTOR table with correct Primary Key*

*Award **[1]** for PatID foreign key in the VISIT table*

*Award **[1]** for DocID foreign key in the VISIT table*

*Award **[1]** for VisitID or PatID/DocID/VisitDate composite key in the VISIT table*

```
PATIENT(PatID, PatientName, PatientDOB)
DOCTOR(DocID, Doctor)
VISIT(VisitID, PatID, DocID, VistDate, Treatment)
```

4. (a) *Award **[2]** max*

*Award **[1]** per point*

Dataset size/data volume/number of features;
Algorithm/model complexity;
Feature engineering;
Hyperparameter tuning;
Real-time requirements;
Ensemble methods.

(b) *Award **[2]** max*

*Award **[1]** per point, from any section*

Audit: Carry out independent audits to check for algorithmic bias;
Validate: Run validations to ensure that the algorithm is not biased;
Cross-reference diagnosis with human diagnosis;

Public disclosure: Inform patients about the algorithm and its limitations;
Provide informed consent forms for patients to sign;

Committee: Set up an oversight committee to approve the use of machine learning;
Any significant change to algorithms or datasets requires committee approval;
Have a patient representative on the committee;

Section B

5. (a) *Award **[2]** max*

*Award **[1]** for identifying issue and **[1]** for expansion*

Reputational damage;
Clients may lose confidence in a company that has a breach of security;
Clients may use a different hospital/refuse to use that hospital;

Financial fraud for patients;
Patients' credit card numbers/banking data/passwords may result in funds being removed;

Theft of data;
Patients' data is used unethically (e.g., insurance company refusing to insure patients with potential pre-existing medical conditions);

Identity theft;
Having your identity stolen can cause psychological damage; e.g., a person may steal a person's online profile and insult others causing emotional distress;

Data loss causing disruption to hospital services;
Phishing attacks are often designed to steal sensitive data; e.g. doctors lose access to patients records and can't treat them;

Legal consequences;
Company may be sued for leaking sensitive information by patients/companies/government/ombudsman/lose license to practice medicine;

Create vulnerability for additional malware;
Causing further disruption.

(b) *Award **[4]** max*

*Award **[1]** for identifying method and **[1]** for expansion*

Packet filtering: a firewall inspects incoming data packets;
Blocking or allowing packets based on predefined rules (e.g., content, location sent from);

Stateful inspection: Monitors active connections;
Only packets that come from an established connection or session are accepted;

Proxy service: Firewalls can act as a proxy inspecting and forwarding packets;
Prevents direct access to a system,

Blocking unwanted traffic: Preventing access from harmful sites;
Configured to block data from certain locations or applications;

Network address translation (NAT): Presents a single, public-facing IP address;
Making it harder for attackers to find individual systems;

Application layer filtering: Inspect data up to the application layer;
Blocking specific activities within protocols, like file uploads, instant messaging;

Intrusion detection and prevention: Mechanisms to identify suspicious patterns;
Alerting administrators or actively blocking the suspicious traffic;

Segmentation: Firewalls can be used to segment different parts of the network;
Sensitive data/operations can be kept separate from general use;

Virus and malware scanning: integrated malware scanners;
Detecting malware as soon as it enters the network;

Denial of service protection: limit the rate of requests from a particular IP address;
Helps to mitigate distributed denial-of-service (DDoS) attacks;

User authentication: can be set up to require user authentication;
So that only registered users can access the network resources;

Mark [2] + [2]

(c) *Award **[6] max***

Responses may include some of the following points. The best ones will demonstrate research and give a balanced perspective before presenting a conclusion.

Steps to reduce potential risks

- Implement robust security protocols: TaylorMade Healthcare can establish the following steps.
- Security protocols: Ensure there is a failover system for firewalls, security software, email filtering, and regular software updates.
- Device management: Implement a mobile device management (MDM) solution to control which devices can access company data, enforce security policies, and remotely wipe data if necessary.
- VPN and secure connections: Ensure that all devices connecting to the company network do so through a secure connection like a VPN.
- Regular audits: Periodically check and verify the security status of all devices that connect to the network.
- Data segregation: Use applications that can segregate corporate data from personal data on the device.
- User awareness training: Educating staff about potential threats, such as phishing scams and ransomware attacks, can prevent many attacks. Candidates should be aware that human error remains a significant challenge. TaylorMade Healthcare should have clearly defined procedures and protocols for new staff and departing staff, such as not connecting to open Wi-Fi networks and not downloading untrusted apps.
- Employees should be made aware of the damage caused by malware.

Risks for bring your own device (BYOD) policy

- Device diversity: A variety of device types, operating systems, and software versions can make standardized security measures challenging.
- Uncontrolled environments: Personal devices might connect to insecure networks, increasing the risk of exposure to threats.
- Data leakage: There is a risk of sensitive organizational data being stored alongside personal data and potentially being shared inadvertently.
- Loss or theft: Personal devices can easily be lost or stolen, risking the exposure of any stored organizational data.

Ransomware

- A type of malware that infiltrates and encrypts data, holding it hostage until a ransom is paid.
- Typically infiltrates through phishing emails, drive-by download, malvertising, exploit kits, removable media, and remote desktop protocols (RDP) exploits.
- Once activated, the ransomware payload runs, sometimes with a delay or when triggered by specific user actions.
- The ransomware scans the network for specific file types and begins to encrypt them. The encryption keys are typically held only by the attacker.
- After encryption, the ransomware communicates with a command and control (C&C) server operated by the attacker, signalling the attack has been successful.
- The victim is presented with a time-dependent demand for payment, typically in cryptocurrency to maintain the attacker's anonymity.
- If the victim pays, the attacker may send the decryption tool or key.
- Some ransomware strains delete themselves and activities after the attack, making it harder for authorities to trace them.

Rootkit attacks:

- A stealthy type of malware designed to remain hidden, allowing attackers to maintain prolonged unauthorized access.
- Typically infiltrates systems through exploiting vulnerabilities, bundled with other malware, or malicious software installations.
- Once installed, the rootkit may modify system binaries or install itself as a kernel module or driver.
- The rootkit malware can intercept system calls and manipulate output to hide its presence.
- Communicates with a command and control (C&C) server, allowing the attacker to remotely execute commands and steal data.
- Advanced rootkits can patch system updates or alter firmware, persisting even if the operating system is reinstalled.
- Many rootkits include functions to download and install additional malware, transforming the compromised machine into a bot.
- Signature-based approaches often fail to identify rootkits so behavioural-based detection is required to identify and remove them.

Conclusion
A reasoned conclusion that expresses an opinion and is supported by evidence.

Marks	Level descriptor
No marks	• No knowledge or understanding of the relevant issues and concepts. • No use of appropriate terminology.
Basic 1–2 marks	• Minimal knowledge and understanding of the relevant issues. • Minimal use of appropriate terminology. • The answer may be little more than a list. • No reference is made to the case study or independent research.
Adequate 3–4 marks	• A descriptive response with limited knowledge and/or understanding. • A limited use of appropriate terminology. • There is some analysis. • There is evidence of some research.
Competent 5–6 marks	• Knowledge and understanding of the related issues and/or concepts. • Uses terminology appropriately in places. • There is evidence of analysis. • There is evidence of research. • There is conclusion

Diploma Programme
Programme du diplôme
Programa del Diploma

Computer science
Standard level
Paper 2

Specimen paper

1 hour 15 minutes

Instructions to candidates

- Do not open this examination paper until instructed to do so.
- Answer all of the questions from **one** of the options.
- The maximum mark for this examination paper is **[50 marks]**.

Option	Questions
Option A — Java	1 – 3
Option B — Python	4 – 6

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Option A — Java

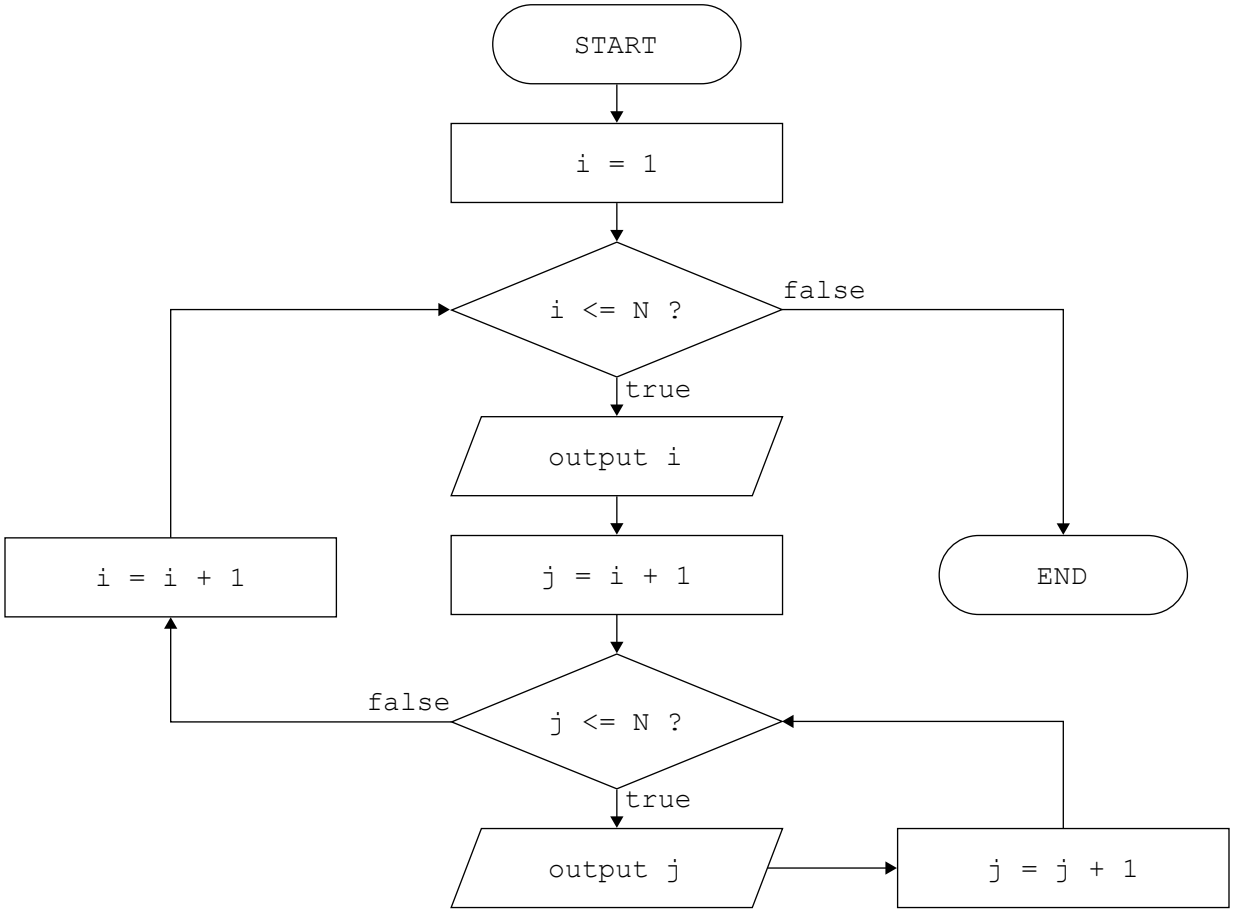
Answer **all** questions. All code that forms part of your responses must be written in Java.

1. Algorithms can be represented by flowcharts.

(a) State **one** reason why a flowchart could be used to represent an algorithm. [1]

Consider the flowchart shown in **Figure 1**.

Figure 1: Flowchart for an algorithm



(b) Copy and complete the trace table formed by the processing of this flowchart, where `N = 3`. [3]

i	i <= N?	j	j <= N ?	output
1				

(Option A continues on the following page)

(Option A, question 1 continued)

Big O notation is used to indicate the time complexity of an algorithm.

(c) State the Big O notation for the algorithm shown in **Figure 1**. [1]

The text file, employees, contains the following information about each employee in a company:

- Surname
- First name
- Role

Each item of data is stored in a new line. Details of the first three employees are shown:

Roberts
John
Sales
Saied
Abdullah
Manager
Sanchez
Pedro
IT manager
…

An algorithm reads the data from the file, employees, line by line. The algorithm writes only the surname of each employee into a new text file, surnames. Both files are currently closed.

(d) Describe the steps that would be followed by this algorithm. [5]

(Option A continues on page 5)

Blank page

(Option A, question 1 continued)

Stacks are a data structure frequently used in computer science to represent real-world scenarios.

Table 1 shows the fundamental operations that operate on a stack.

Table 1: Fundamental operations that operate on a stack

`push()`	Inserts an element onto the top of the stack.
`pop()`	Removes an element from the top of the stack.
`peek()`	Returns the top element of the stack without removing it from the stack.
`isEmpty()`	Returns true if the stack is empty, else it returns false.

(e) Sketch the stack resulting from the following operations.

```
push(1)
push(2)
push(9)
peek()
pop()
push(8)
push(10)
pop()
push(6)
```

You should clearly indicate the top of the stack. [2]

A new stack has been created and filled with elements.

(f) Using only the stack operations shown in **Table 1**, describe the steps that will locate **and** print only the **first** element that was pushed onto the new stack. [3]

(Option A continues on the following page)

(Option A continued)

2. A theatre stores the names and email addresses of its clients in two separate one-dimensional (1D) arrays, where each client's name and email address have corresponding indices. The two arrays are shown in **Table 2**.

Table 2: Name and email arrays

Name array

Index	Name
0	Maria Gonzalez
1	Olga Sarkozky
2	Julia Brown
3	…

Email array

Index	Email address
0	magonzalez24@example.net
1	osarkozky12@one2seven.org
2	jbrown23@example.com
3	…

(a) Explain why arrays have been chosen to hold this data instead of separate variables. [3]

A method has been constructed that will change each name so that the surname appears before the first name. For example, "Maria Gonzalez" will become "Gonzalez Maria".

(b) Describe the steps involved in this method.

Assume that there is only one space between each first name and surname. [4]

Consider the following two-dimensional (2D) array, `MAT`.

```
int[][] MAT = {{2, 7, 1}, {8, 9 ,0}}
```

(c) State the code that will identify the data item `0` in `MAT`. [1]

(Option A continues on the following page)

(Option A, question 2 continued)

The following code operates on `MAT`:

```
int[][] MAT = {{2, 7, 1}, {8, 9 ,0}};
int a = 0;
int b = 0;
for (int row = 0; row < 2; row++)
{
    for (int col = 0; col < 3; col++)
    {
        a = a + 1;
        int item = MAT[row][col];
        if (item%2 != 0)
            b = b + item;
    }
}
System.out.println ("The value of a is : " + a);
System.out.println ("The value of b is : " + b);
```

(d) State the output from this code. [2]

The theatre employs a 2D array to indicate the status of each seat for a performance. `true` indicates that the seat has been sold for that performance. `false` indicates that it has not been sold.

The 2D array `seats` is used to represent the seating plan in the theatre. The array contains 40 rows and 30 columns.

For example, `seats[2][3] = false` shows that the seat in row 2, column 3 of the 2D array, `seats`, has not been sold.

After each performance, the sales data is analysed with a program that uses the method `least()`. This accepts the 2D array seats and outputs the row for which the lowest number of seats were sold.

(e) Construct the method `least()`.

Assume only one row matches this condition. [5]

(Option A continues on the following page)

(Option A continued)

3. A company employs a number of salespeople to sell its goods.

Details of the sales made by each employee for a year are stored in the `SalesPerson` object for that employee.

At the end of each month, the total sales for each employee are added to the array `sales` in the `SalesPerson` object. At the end of the year, the array will contain 12 entries, one for each month of the year.

Bonuses are paid to employees at the end of each year.

The `SalesPerson` class is partially shown:

```
public class SalesPerson
{
    static int bonus1 = … ;          // value to be added
    static int bonus2 = … ;          // value to be added
    private String id;               // unique id for each employee
    double[] sales = new double[12]; // sales figures for each month in a year

    public SalesPerson(String id)
    {
        this.id = id;
    }

    public String getId() // returns employee's id
    {
        return this.id;
    }

    public double calculateTotal() // returns the total sales figure for
the year
    {
        // code missing
    }

    public double calculateHighest() // returns highest monthly total in
the year
    {
         // code missing
    }

    public void addSales(…) // adds the sales for a month to the sales array
    {
        // code missing
    }
}
```

(Option A continues on the following page)

(Option A, question 3 continued)

The driver code can be used to instantiate new `SalesPerson` objects. For example:

```
SalesPerson employee1 = new SalesPerson ("A6457098");
```

(a) List the operations that would be carried out by the constructor when this line of code is run. [4]

An object can have different types of attributes.

(b) Outline **one** difference in the way in which the `bonus1` and `id` attributes are applied in the `SalesPerson` class. [2]

The variable `id` has been identified as being a private attribute of the class.

(c) Outline **two** consequences of declaring an attribute as private. [2]

At the end of each month, the `SalesPerson` method, `addSales()`, adds the total sales for that month to the sales array for each employee.

The parameters passed to this method are the total sales for that month and the number representing that month, where 0 corresponds to January.

For example:

```
employee1.addSales(30000,2); // this adds 30000 to the sales array for
the month of March for employee1.
```

(d) Construct the method `addSales()`. [2]

The `SalesPerson` method `calculateHighest()` returns the highest monthly sales figure by that employee during the year.

(e) Construct the method `calculateHighest()`. [5]

(Option A continues on page 11)

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(Option A, question 3 continued)

The company has 10 employees. An array, `employee`, of size 10 holds `SalesPerson` objects, one for each employee.

End-of-year bonuses, shown in the `SalesPerson` class, are calculated for each employee as follows:

- `bonus1` is paid if the total sales for the year are more than 300 000.
- `bonus2` is paid if the sales for any one month are more than 20 000.

The driver code includes the method `calculateBonus()`.

For each employee, this method:

- calculates the bonuses
- outputs their employee ID and their total bonus.

(f) Construct the method `calculateBonus()`.

Use any of the methods listed in the `SalesPerson` class. [5]

End of Option A

Option B — Python

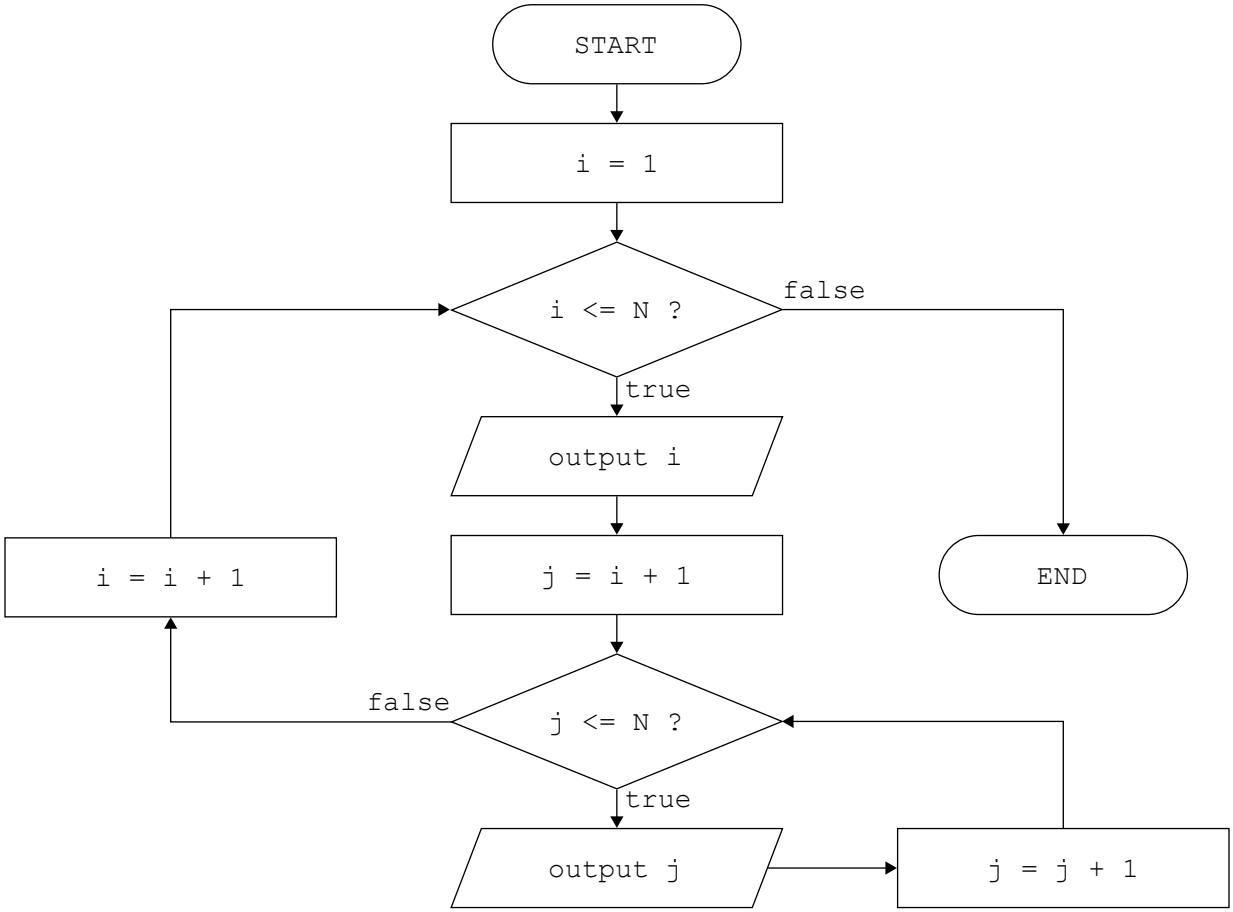
Answer **all** questions. All code that forms part of your responses must be written in Python.

4. Algorithms can be represented by flowcharts.

(a) State **one** reason why a flowchart could be used to represent an algorithm. [1]

Consider the flowchart shown in **Figure 2**.

Figure 2: Flowchart for an algorithm



(b) Copy and complete the trace table formed by the processing of this flowchart, where `N = 3`. [3]

i	i <= N?	j	j <= N ?	output
1				

(Option B continues on the following page)

(Option B, question 4 continued)

Big O notation is used to indicate the time complexity of an algorithm.

(c) State the Big O notation for the algorithm shown in **Figure 2**. [1]

The text file, employees, contains the following information about each employee in a company:

- Surname
- First name
- Role

Each item of data is stored in a new line. Details of the first three employees are shown:

Roberts
John
Sales
Saied
Abdullah
Manager
Sanchez
Pedro
IT manager
…

An algorithm reads the data from the file, employees, line by line. The algorithm writes only the surname of each employee into a new text file, surnames. Both files are currently closed.

(d) Describe the steps that would be followed by this algorithm. [5]

(Option B continues on page 15)

Blank page

(Option B, question 4 continued)

Stacks are a data structure frequently used in computer science to represent real-world scenarios.

Table 3 shows the fundamental operations that operate on a stack.

Table 3: Fundamental operations that operate on a stack

`push()`	Inserts an element onto the top of the stack.
`pop()`	Removes an element from the top of the stack.
`peek()`	Returns the top element of the stack without removing it from the stack.
`isEmpty()`	Returns true if the stack is empty, else it returns false.

(e) Sketch the stack resulting from the following operations.

```
push(1)
push(2)
push(9)
peek()
pop()
push(8)
push(10)
pop()
push(6)
```

You should clearly indicate the top of the stack. [2]

A new stack has been created and filled with elements.

(f) Using only the stack operations shown in **Table 3**, describe the steps that will locate **and** print only the **first** element that was pushed onto the new stack. [3]

(Option B continues on the following page)

(Option B continued)

5. A theatre stores the names and email addresses of its clients in two separate one-dimensional (1D) lists, where each client's name and email address have corresponding indices. The two lists are shown in **Table 4**.

Table 4: Name and email lists

Name list

Index	Name
0	Maria Gonzalez
1	Olga Sarkozky
2	Julia Brown
3	…

Email list

Index	Email address
0	magonzalez24@example.net
1	osarkozky12@one2seven.org
2	jbrown23@example.com
3	…

(a) Explain why lists have been chosen to hold this data instead of separate variables. [3]

A function has been constructed that will change each name so that the surname appears before the first name. For example, "Maria Gonzalez" will become "Gonzalez Maria".

(b) Describe the steps involved in this function.

Assume that there is only one space between each first name and surname. [4]

Consider the following two-dimensional (2D) list, `MAT`.

```
MAT = [[2, 7, 1],
       [8, 9 ,0]]
```

(c) State the code that will identify the data item `0` in `MAT`. [1]

(Option B continues on the following page)

(Option B, question 5 continued)

The following code operates on `MAT`:

```
a = 0
b = 0
for row in range (0, 2):
    for col in range (0, 3):
        a = a + 1
        item = MAT[row][col]
        if item%2 != 0:
            b = b + item
print ('The value of a is : ',a)
print ('The value of b is : ',b)
```

(d) State the output from this code. [2]

The theatre employs a 2D list to indicate the status of each seat for a performance. `True` indicates that the seat has been sold for that performance. `False` indicates that it has not been sold.

The 2D list `seats` is used to represent the seating plan in the theatre. The list contains 40 rows and 30 columns.

For example, `seats[2][3] = False` shows that the seat in row 2, column 3 of the 2D list, `seats`, has not been sold.

After each performance, the sales data is analysed.

One program uses the function `least()`. This accepts the 2D list `seats` and outputs the row for which the lowest number of seats were sold.

(e) Construct the function `least()`.

Assume only one row matches this condition. [5]

(Option B continues on the following page)

(Option B continued)

6. A company employs a number of salespeople to sell its goods.

Details of the sales made by each employee for a year are stored in the `SalesPerson` object for that employee.

At the end of each month, the total sales for each employee are added to the list `self._sales` in the `SalesPerson` object. At the end of the year, the list will contain 12 entries, one for each month of the year.

Bonuses are paid to employees at the end of each year.

The `SalesPerson` class is partially shown:

```
class SalesPerson():

    bonus1 = …          # value to be added
    bonus2 = …          # value to be added


    def __init__(self, id):
        self.__id = id          # employee id
        self.__sales = []       # holds the monthly sales figures

    def get_id(self):           # returns employee's id
        return self._id

    def calculate_total(…):     # returns the total sales figure for the year
        # code missing

    def calculate_highest(…): # returns highest monthly total in the year
        # code missing

    def add_sales(…):           # adds the sales for a month to the sales list
        # code missing
```

(Option B continues on the following page)

(Option B, question 6 continued)

The driver code can be used to instantiate new `SalesPerson` objects. For example:

```
employee1 = SalesPerson ("A6457098")
```

(a) List the operations that would be carried out by the constructor when this line of code is run. [4]

An object can have different types of attributes.

(b) Outline **one** difference in the way in which the `bonus1` and `self._id` attributes are applied in the `SalesPerson` class. [2]

The underscore character in `self._id = id` identifies the variable as being a private attribute of the class.

(c) Outline **two** consequences of declaring an attribute as private. [2]

At the end of each month, the `SalesPerson` method, `add_sales()`, adds the total sales for that month to the sales list for each employee.

For example:

```
employee1.add_sales(30000)  # this adds 30000 to the sales list
for employee1.
```

(d) Construct the method `add_sales()`. [2]

The `SalesPerson` method `calculate_highest()` returns the highest monthly sales figure by that employee during the year.

(e) Construct the method `calculate_highest()`.

The `max()` function must not be used. [5]

(Option B continues on page 21)

Blank page

(Option B, question 6 continued)

The company has 10 employees. A list, `employee`, of size 10 holds `SalesPerson` objects, one for each employee.

End-of-year bonuses, shown in the `SalesPerson` class, are calculated for each employee as follows:

- `bonus1` is paid if the total sales for the year are more than 300 000.
- `bonus2` is paid if the sales for any one month are more than 20 000.

The driver code includes the function `calculate_bonus()`.

For each employee, this function:

- calculates the bonuses
- outputs their employee ID and their total bonus.

(f) Construct the function `def calculate_bonus()`.

Use any of the methods listed in the `SalesPerson` class. [5]

End of Option B

Diploma Programme
Programme du diplôme
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Markscheme

Specimen paper

Computer science

Standard level

Paper 2

17 pages

International Baccalaureate®
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General marking instructions

1. Follow the markscheme provided, award only whole marks and mark only in **RED**.

2. Make sure that the question you are about to mark is highlighted in the mark panel on the right-hand side of the screen.

3. Where a mark is awarded, a tick/check (✓) **must** be placed in the text at the **precise point** where it becomes clear that the candidate deserves the mark. **One tick to be shown for each mark awarded**.

4. Sometimes, careful consideration is required to decide whether or not to award a mark. In these cases use RM™ Assessor annotations to support your decision. You are encouraged to write comments where it helps clarity, especially for re-marking purposes. Use a text box for these additional comments. It should be remembered that the script may be returned to the candidate.

5. Personal codes/notations are unacceptable.

6. Where an answer to a part question is worth no marks but the candidate has attempted the part question, enter a zero in the mark panel on the right-hand side of the screen. Where an answer to a part question is worth no marks because the candidate has not attempted the part question, enter an “NR” in the mark panel on the right-hand side of the screen.

7. Please ensure you check all scanned pages. The candidate may have answered more than one option.

8. **If a candidate has attempted more than one Option** within a paper, mark all the candidate’s work. RM™ Assessor will only award the marks for the higher scoring Option. Once all the work the candidate has attempted has been marked, please click “COMPLETE”; all the other questions from the other Options will auto complete to “NR” for “no response”.

9. Ensure that you have viewed **every** page including any additional sheets. Please ensure that you stamp “SEEN” on any page that contains no other annotation.

10. A mark should not be awarded where there is contradiction within an answer. Make a comment to this effect using a text box or the “CON” stamp.

Subject details: Computer Science SL Paper 2 markscheme

Mark allocation

Candidates are required to answer **all** questions in **one** Option. Total 50 marks.

General

A markscheme often has more specific points worthy of a mark than the total allows. This is intentional. Do not award more than the maximum marks allowed for that part of a question.

When deciding upon alternative answers by candidates to those given in the markscheme, consider the following points:

- Each statement worth one point has a separate line and the end is signified by means of a semi-colon (;).
- An alternative answer or wording is indicated in the markscheme by a "/"; either wording can be accepted.
- Words in (…) in the markscheme are not necessary to gain the mark.
- If the candidate's answer has the same meaning or can be clearly interpreted as being the same as that in the markscheme then award the mark.
- Mark positively. Give candidates credit for what they have achieved and for what they have got correct, rather than penalizing them for what they have not achieved or what they have got wrong.
- Remember that many candidates are writing in a second language; be forgiving of minor linguistic slips. In this subject effective communication is more important than grammatical accuracy.
- Occasionally, a part of a question may require a calculation whose answer is required for subsequent parts. If an error is made in the first part then it should be penalized. However, if the incorrect answer is used correctly in subsequent parts then **follow through** marks should be awarded. Indicate this with "**FT**".

General guidance

Issue	Guidance
Answering more than the quantity of responses prescribed in the questions	• In the case of an "identify" question, read all answers and mark positively up to the maximum marks. Disregard incorrect answers. • In the case of a "describe" question, which asks for a certain number of facts *eg* "describe two kinds", mark the first two correct answers. This could include two descriptions, one description and one identification, or two identifications. • In the case of an "explain" question, which asks for a specified number of explanations *eg* "explain two reasons …", mark the first two correct answers. This could include two full explanations, one explanation, one partial explanation *etc*.

Option A — Java

1. (a) *Award **[1 max]***

because it is in diagrammatic form;
it is program agnostic;
no complicated code / syntax to follow;

(b) *Award **[3]***

correct values of i;
correct values of j;
correct output;

i	i < = N ?	j	j < = N ?	output
1	TRUE			1
		2	TRUE	2
		3	TRUE	3
		4	FALSE	
2	TRUE			2
		3	TRUE	3
		4	FALSE	
3	TRUE	4	FALSE	3
4	FALSE			

(c) *Award **[1]**. Note: the algorithm shows a nested loop.*

On^2;

(d) *Award **[5 max]** as follows*

opening *employees* file for reading / default mode;
open *surnames* file for writing;
looping through *employees* file;
use of a counter or equivalent;
writing surnames only to *surnames* file;
closing both files;

*Note: Award **[1]** for first 2 marking points if files are opened without indicating access modes.*

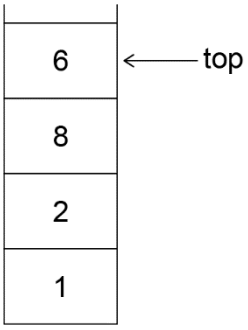
Algorithm is:

```
open employees file for reading / in default mode
open surnames file for writing
set counter to 0
repeat until end of employees file
    read next line from employees file
    if counter%3 = 0 *
        write line to surnames file
    increment counter
close both files
```

* there are alternatives ways of using counter

(e) *Award **[2]**.*

correct elements in stack **[1]**;
correct indication of top **[1]**;



(f) *Award **[3]** as follows:*

*Award **[1]** for each correct line (or equivalent).*

repeat until `isEmpty` is `true` / while `isEmpty` is `false`;
pop an element and save its value/save in a variable/push it onto a new stack (stack2)
print the last value saved/print the variable/pop stack2 and print value

Note: Answer must demonstrate an algorithmic approach.

2. (a) *Award **[3 max]***

It is easier to work with one single structure than several individual ones;
Individual elements that have the same value can be quickly accessed via their indices;
Iteration through arrays is more efficient / Methods associated with arrays can make operations more efficient;

(b) *Award **[4 max]***

repeat for each entry/index in the `Name` array
    identify the position of the space in each name
    save the substring to the left as s1 and the substring to the right as s2
    save the new entry as s2+space+s1

(c) *Award **[1]***

```
MAT[1][2];
```

(d) *Award **[1]** each;*

The value of a is: 6
The value of b is: 17

(e) *Award **[5 max]** as follows:*

*Award **[1]** for initial variables correctly set;*
*Award **[1]** for both loops;*
*Award **[1]** for checking seat for* `False`*;*
*Award **[1]** for updating row value;*
*Award **[1]** for resetting* `tempTotal`*;*
*Award **[1]** for printing the correct row;*

```
public void least(boolean[][]seats)
{
    int falseTotal = 0;
    int leastRow = 0;

    for (int x = 0; x < 40; x++)
    {
        int tempTotal = 0;
        for (int y = 0; y < 30; y++)
        {
         if (seats[x][y] == false)       // if(!seats[x][y] )
            tempTotal = tempTotal + 1;
        }
        if (tempTotal > falseTotal)
        {
            leastRow = x;
            falseTotal = tempTotal;
        }
    }
    System.out.println ("Row with least seats sold is " + leastRow);
}
```

3. (a) *Award **[4 max]***

creates / assigns memory to a new instance of the `SalesPerson` class;
identified by the variable employee1;
initialises id with value “A6457098”;
declares (empty) sales array;

(b) *Award **[2 max]***

`bonus1` belongs to the class / it will remain the same for all instances of the class / has the same memory for all instances;
`id` belongs to the object/instance / it will be different for each instance / be assigned different memory for each instance;

`id` is declared inside the constructor method;
`bonus1` is declared inside the class definition but outside any of the instance methods and constructor;

`id` gets created when an instance of the class is created;
`bonus1` is created when the program begins to execute;

changes made to instance variable id through one object will not reflect in another object;
changes made in the class variable bonus1 will reflect in all objects;

(c) *Award **[2 max]***

provides greater security for the values stored in these variables/for the code;
restricts access to such variables <u>from outside of the class</u> /only allows access through public / getter, setter methods;
(helps to) prevents <u>accidental/unintended</u> changes to the variables (from outside of the class);

(d) *Award **[2 max]***
*Award **[1]** for parameters;*
*Award **[1]** for adding the sales figure to the array;*

```
public void addSales(double amount,int month)
{
   sales[month] = amount;
}
```

(e) *Award **[5 max]***

*Award **[1]** for a correct loop;*
*Award **[1]** for initializing highest;*
*Award **[1]** for the comparison;*
*Award **[1]** for updating highest;*
*Award **[1]** for returning highest;*

```
public double calculateHighest()
{
    double highest = 0;
    for (int x = 0; x < 12; x++)
    {
        if (this.sales[x] > highest)
            highest = this.sales[x];
    }
    return highest;
}
```

(f) *Award **[5 max]***

*Award **[1]** for a loop <u>and</u> initialising <u>and</u> updating bonus variable;*
*Award **[1]** for a correct comparison using `calculateTotal` method;*
*Award **[1]** for a correct comparison using `calculateHighest` method;*
*Award **[1]** for a correct identification of `bonus1` and `bonus2`;*
*Award **[1]** for prints out the id (using getter) and bonus (allow values only);*

```
public void bonus(SalesPerson[] employee)
{
    for (int x = 0; x < 10; x++)
    {
        int bonus = 0;
        if (employee[x].calculateTotal() > 300000)
            bonus = bonus + SalesPerson.bonus1;
        if (employee[x].calculateHighest() > 20000)
            bonus = bonus + SalesPerson.bonus2;
        System.out.println ("ID is " + employee[x].getId());
        System.out.println ("Bonus is " + bonus);
    }
}
```

Option B — Python

4. (a) *Award **[1 max]***

because it is in diagrammatic form;
it is program agnostic;
no complicated code / syntax to follow;

(b) *Award **[3]***

correct values of i;
correct values j;
correct output;

`i`	`i < = N ?`	`j`	`j < = N ?`	`output`
`1`	TRUE			1
		2	TRUE	2
		3	TRUE	3
		4	FALSE	
2	TRUE			2
		3	TRUE	3
		4	FALSE	
3	TRUE	4	FALSE	3
4	FALSE			

(c) *Award **[1]**. Note: the algorithm shows a nested loop.*

On^2;

(d) *Award **[5 max]** as follows*

opening *employees* file for reading / default mode;
open *surnames* file for writing;
looping through *employees* file;
use of a counter or equivalent;
writing surnames only to *surnames* file;
closing both files;

*Note: Award **[1]** for first 2 marking points if files are opened without indicating access modes*

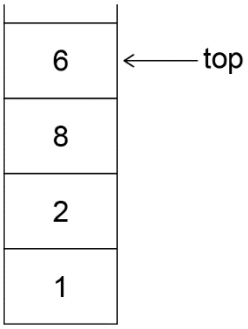
Algorithm is:

```
open employees file for reading / in default mode
open surnames file for writing
set counter to 0
repeat until end of employees file
    read next line from employees file
    if counter%3 = 0 *
        write line to surnames file
    increment counter
close both files
```

* there are alternatives ways of using counter

(e) *Award **[2]**.*

correct elements in stack **[1]**;
correct indication of top **[1]**;



(f) *Award **[3]** as follows:*

*Award **[1]** for each correct line (or equivalent).*

repeat until `isEmpty` is `true` / while `isEmpty` is `false`;
pop an element and save its value/save in a variable/push it onto a new stack (stack2);
print the last value saved/print the variable/pop stack2 and print value;

Note: Answer must demonstrate an algorithmic approach.

5. (a) *Award **[3 max]***

It is easier to work with one single structure than several individual ones;
Individual elements that have the same value can be quickly accessed via their indices;
Iteration through arrays is more efficient / Methods associated with arrays can make operations more efficient;

(b) *Award **[4 max]** as follows:*

repeat for each entry/index in the `Name` list
identify the position of the space in each name
save the substring to the left as s1 and the substring to the right as s2
save the new entry as s2+space+s1

repeat for each entry/index in the `Name` list
use the `split()` function
to separate the 2 strings into 2 list values (s1 and s2)
save the new entry as s2+space+s1

(c) *Award **[1]***

```
MAT[1][2];
```

(d) *Award **[1]** for each:*

The value of a is: 6
The value of b is: 17

(e) *Award **[5 max]** as follows:*

*Award **[1]** for initial variables correctly set;*
*Award **[1]** for both loops;*
*Award **[1]** for checking seat for* `False`*;*
*Award **[1]** for updating row value;*
*Award **[1]** for resetting* `temp_total`*;*
*Award **[1]** for printing the correct row;*

```
def least(seats):

    false_total = 0
    least_row = 0

    for x in range(40):
        temp_total = 0
        for y in range (30):
            if seats[x][y] == False:
                temp_total = temp_total + 1
        if temp_total > false_total:
            least_row = x
            false_total = temp_total

    print ("Row with least seats sold is ",least_row)
```

6. (a) *Award **[4 max]***

creates / assigns memory to a new instance of the `SalesPerson` class;
identified by the variable employee1;
initialises id with value “A6457098”;
declares (empty) sales list;

(b) *Award **[2 max]***

`bonus1` belongs to the class / it will remain the same for all instances of the class / has the same memory for all instances;
`self_id` belongs to the object/instance / it will be different for each instance / be assigned different memory for each instance;

`self._id` is declared inside the constructor/ the `__init__()` method;
`bonus1` is declared inside the class definition but outside any of the instance methods and constructor;

`self.id` variable gets created when an instance of the class is;
`bonus1` is created when the program begins to execute;

changes made to instance variable `self._id` through one object will not reflect in another object; changes made in the class variable `bonus1` will reflect in all objects;

(c) *Award **[2 max]***

provides greater security for the values stored in these variables/for the code;
restricts access to such variables <u>from outside of the class</u> /only allows access through public / getter, setter methods;
(helps to) prevents <u>accidental/unintended</u> changes to the variables (from outside of the class);

(d) *Award **[2 max]***
*Award **[1]** for parameters;*
*Award **[1]** for adding the sales figure to the list;*

```
def add_sales(self, amount):
    self._sales.append(amount)

def add_sales(self, amount):
    self._sales = self._sales + [amount]
```

(e) *Award **[5 max]***
*Award **[1]** for a correct loop;*
*Award **[1]** for initializing highest;*
*Award **[1]** for the comparison;*
*Award **[1]** for updating highest;*
*Award **[1]** for returning highest;*

```
def calculate_highest(self):
    highest = 0
    for x in range (12):                # ... range (0,12)
        if self._sales[x] > highest:
            highest = self._sales[x]
    return highest
```

```
def calculate_highest(self):           l
    highest = 0
    for x in range (len(self._sales)):
        if self._sales[x] > highest:
            highest = self._sales[x]
    return highest
```

```
def calculate_highest(self):
    highest = 0
    for sales_figure in self._sales:
        if sales_figure > highest:
            highest = sales_figure
    return highest
```

(f) *Award **[5 max]***

*Award **[1]** for a loop <u>and</u> initialising <u>and</u> updating bonus variable;*
*Award **[1]** for a correct comparison using `calculate_total` method;*
*Award **[1]** for a correct comparison using `calculate_highest` method;*
*Award **[1]** for a correct identification of `bonus1` and `bonus2`;*
*Award **[1]** for prints out the id (using getter) and bonus (allow values only);*

```
def calculate_bonus():
    for x in range(10):        # … range (0,10)    … range
(len(self._sales))

        bonus = 0
        if employee[x].calculate_total() > 300000:
            bonus = bonus + SalesPerson.bonus1
        if employee[x].calculate_highest() > 20000:
            bonus = bonus + SalesPerson.bonus2
        print (f"Employee {employee[x].get_id()} has bonus ", bonus).
```

Diploma Programme
Programme du diplôme
Programa del Diploma

Computer science
Case study: Preventing network security attacks

Specimen paper

Instructions for candidates

- Case study booklet required for higher level and standard level paper 1 computer science examinations.

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Introduction

The integration of technology in healthcare has transformed how medical services are delivered. However, with this increased use of technology, healthcare institutions like private hospitals have become attractive targets for cybercriminals and therefore face significant security challenges that could have devastating impacts. This case study highlights the interventions used to prevent security incidents by one private hospital, TaylorMade Healthcare, and explores some network security practices.

Background

TaylorMade Healthcare uses high-speed internet to operate its network infrastructure, which includes computers, servers, switches, routers, wireless access points, and *firewalls*. These devices and systems are connected internally and to the internet, enabling staff to access medical records, communicate with other users and patients, exchange information, and transfer data. The devices and systems work cohesively and are vital for the hospital's smooth operation, including patient care, data management, and resource handling.

TaylorMade Healthcare stores vast amounts of sensitive data, such as patient health records, financial information, and confidential documents, which it must safeguard to maintain trust and confidence among its patients, staff, and stakeholders.

The hospital's director acknowledges the importance of securing the network and data against risks like *malware* infections, *phishing* attacks, and unauthorized access. The director also knows that it is critical to implement effective security measures to protect the hospital's network infrastructure and data.

Bring-your-own-device (BYOD) policy

TaylorMade Healthcare has adopted a bring-your-own-device (BYOD) policy that permits staff to use personal electronic devices for work tasks. Although a BYOD policy can provide many benefits, such as increased flexibility and convenience, it also has several weaknesses.

Ransomware and *rootkit* attacks

TaylorMade Healthcare recently suffered a ransomware attack that targeted its network infrastructure. The attack was launched through a phishing email that appeared to be from a trusted source. When an unsuspecting staff member clicked on the link in the email, the ransomware gained access to the hospital's network and quickly spread, encrypting sensitive data and rendering it inaccessible.

The consequences of the attack were severe, leading to significant data loss and system downtime. The ransomware attack encrypted patients' data and prevented staff from accessing patient records for several days. This led to substantial disruptions.

TaylorMade Healthcare's reputation has suffered due to the attack, with patients expressing concern about the security of their data. There was also a significant financial impact, with TaylorMade Healthcare paying the attackers a ransom to regain access to its encrypted data.

Another notable concern is a rootkit attack, which uses malicious software to hide itself within an operating system to gain administrative-level access to the network. Rootkits are notoriously difficult to remove.

Interventions

"Dealing with the ransomware attack was a wake-up call for our entire team," said Jane Wang, Head of Cybersecurity. "We realized the importance of being proactive and creating a culture of security awareness," she added, "and we also realized that a comprehensive security strategy is essential for protecting sensitive data and networks."

Since the attack, TaylorMade Healthcare has identified several measures to prevent future attacks:

- Data backup and *disaster recovery plans*
- *Email filtering*
- *Failover systems*
- Firewall implementation
- Network segmentation and *access control*
- Regular software and security updates
- *Third-party services*
- *User awareness training*

Implementing network segmentation and access control reduced the risk of cyber-attacks and improved the security of TaylorMade Healthcare's network. These and the other interventions have contributed to reducing the effects of security breaches, ensuring the confidentiality and integrity of sensitive information. These safeguards have also minimized system downtime and strengthened data protection, which is vital for the hospital's uninterrupted functioning.

Challenges faced

TaylorMade Healthcare's goal is to build resilience to address the following challenges.

Standard level and higher level:

- Develop strategies to create an awareness of cybersecurity, especially ransomware and rootkit attacks.
- Consider the benefits of a bring-your-own-device (BYOD) policy while preparing for its security weaknesses.

Higher level only:

- Explore the viability of network segmentation and access control.
- Develop procedures to restore operations in case of a breach.

Additional terminology

Access control
Bring your own device (BYOD)
Cybercriminal
Disaster recovery plan
Email filtering
Failover system
Firewall
Malware
Phishing
Ransomware
Rootkit attack
Third-party services
User awareness training

Some companies, products, or individuals named in this case study are fictitious and any similarities with actual entities are purely coincidental.